

Status

HelixTranslate — Feature Status

Revision: 16 **Last modified:** 2026-06-16T19:05:00Z **Authority:** Derived from docs/features/.feature_inventory_raw.md (Revision 1). Per §11.4.45 (captured-evidence Status doc), §11.4.44 (revision header), §11.4.56 (two-audience summary → Status_Summary.md), §11.4.60 (always-sync), §11.4.6 (no-guessing — every status reflects what the inventory actually found in source). **Scope:** Every one of the 417 inventoried features across CLI, API, Web, gRPC, Library, Submodule and Infra. helix_qa deliberately not included.

Anti-bluff note (§11.4.6 / §11.4.2 / §11.4.107): Video-confirmation is set to a real file **only** where a recording exists AND that recording genuinely demonstrates the feature working (each cited .mp4 was ffprobe-verified for non-zero duration+frames and content-verified against its claimed behaviour — e.g. real DeepSeek output in the target language, real gRPC translated text, live WebSocket count change). **66 feature rows are video-confirmed** (Wave 9 / Wave 8 / Wave 7 of 2026-06-16 added **12 net-new** (4 Wave 7 + 5 Wave 8 + 3 Wave 9), each a real input → real LLM (default in-process bridge, NOT -use-verifier) → real output clip, window-scoped (§11.4.154), helix_translate--prefixed (§11.4.155), frame-content-read so source-frame ≠ output-frame per §11.4.107. **Wave 7 (4): EPUB→EPUB round-trip EN→ES** (“The old man walked along the shore at dawn...” → “El anciano caminaba por la orilla al amanecer...”, output EPUB document — helix_translate-epub-to-epub-en-es-wave7-20260616-181058.mp4; **headline re-confirm** — the existing EPUB→EPUB-conversion row was already Confirmed, this clip is an additional distinct-scope confirmation, NOT a new row), **standalone -script Cyrillic→Latin** (pkg/script, NO LLM: “Добар дан свима...ђ...ц” → “Dobar dan svima...đ...dž”, zero Cyrillic remaining — helix_translate-script-cyrillic-to-latin-wave7-20260616-181012.mp4; **headline-only** re-confirm of an already-Confirmed row), **preparation-translator analysis pass** (genre “science fiction”, chars Sarah Mills/Admiral Chen, untranslatable Captain Sarah Mills/Admiral Chen/starship Aurora/New Geneva — helix_translate-preparation-analysis-en-es-wave7-20260616-181123.mp4; **headline-only** re-confirm — the preparation-translator row is already Confirmed) and **REST POST /api/v1/translate/string** (local ./build/server :8080, JSON translated_text = “Las montañas estaban cubiertas de nieve y el río fluía tranquilamente por el valle.”, provider deepseek — a real translation, NOT a session_id — helix_translate-rest-translate-string-en-es-wave7-20260616-181155.mp4; **ROW-FLIPPED** PENDING→Confirmed). **§11.4.138 / §11.4.146 honest finding:** Wave 7

ALSO recorded the **markdown-translator EPUB→MD empty-payload data-loss BUG** (step-3 translate sends empty strings to DeepSeek, the LLM’s “you sent an empty message” boilerplate written AS IF the translation, chapter text lost) — **found broken (wave7) → fixed at source in commit 4f8e581 (this session’s PART-1 Translate-arg data-loss fix) → a re-record is owed before any confirmation**; it was NOT a previously-confirmed row so there is NO demotion, and it is NOT counted as a confirmed net-new.

Wave 8 (5): GROQ provider path EN→ES (groq/llama-3.3-70b-versatile, real Groq API: “La vieja farola se erguía sola sobre el acantilado rocoso...” — helix_translate-groq-en-es-wave8-20260616_182021.mp4; **ROW-FLIPPED** the Groq client library row PENDING→Confirmed), and 4 language-pair confirmations through the deepseek path (**headline-only** — covered by the already-Confirmed generic multi-language-pair coverage, no fabricated per-pair rows): **EN→KO Korean/Hangul** (“새벽에 용이 잠든 숨겨진 성을 찾기 위해 용감한 기사가 고대 숲을 가로질렀다.” — helix_translate-en-ko-korean-wave8-20260616_182059.mp4), **DE→EN reverse** (“The scientist worked all night in his lab...” — helix_translate-de-en-reverse-wave8-20260616_182101.mp4), **FR→ES non-English** (“El anciano marinero contaba historias de tormentas y tesoros ocultos...” — helix_translate-fr-es-pair-wave8-20260616_182111.mp4) and **EN→SR Latin script** (“U srcu planina živela je mudra stara žena...”, all-Latin, no Cyrillic — helix_translate-en-sr-latin-wave8-20260616_182113.mp4). §11.4.6 nit (not a bug): -h provider help omits groq though groq IS fully wired + works end-to-end. **Wave 9 (3): REST POST /api/v1/translate/batch** (live server-w9 :8091 TLS, 3 EN→SR in one request, 0 errors, provider llm-novita — helix_translate-rest-batch-version-wave9-20260616_184823.mp4; **ROW-FLIPPED** PENDING→Confirmed), **REST GET /api/v1/version** (real substantive data: codebase 2.3.0, git_commit c2aa7c8..., go1.26.2, build_time + components map, NOT a bare status — same clip; **headline re-confirm** of the already-Confirmed combined endpoints row) and **preparation→translate two-stage pipeline** (stage-1 real DeepSeek analysis: “short story”/“literary fiction”, chars Captain Marlowe + Eli, 3 untranslatable Stormcrag Point/Captain Marlowe/Eli, 1974 tokens; stage-2 output EPUB preserves exactly those names in Spanish “...El viejo capitán Marlowe había sido farero de Stormcrag Point durante cuarenta años...” — helix_translate-prep-translate-two-stage-wave9-20260616_184823.mp4; **headline-only** — covered by the already-Confirmed preparation-translator row). §11.4.6 / §11.4.124 **honest non-confirmable findings (recorded as gaps, NOT confirmed, never faked)**: pkg/api/server.go alt Server (/api/translate, /api/batch, /api/languages) is **UNWIRED** — no cmd entry instantiates api.NewServer, no runnable binary (did NOT fabricate a runner); and GET /api/v1/providers returns a **static hardcoded list** (openai/anthropic/zhifu/deepseek), NOT substantive real upstream data, so NOT confirmed as substantive per the Wave-9 criterion. **Running tally (§11.4.6 honest arithmetic): 54 (prior authoritative total) + 4 (Wave 7) + 5 (Wave 8) + 3 (Wave 9) = 66**. Of the 12 net-new, **3 are genuine row-flips** (PENDING→Confirmed: POST /api/v1/translate/string, the Groq client library row, POST /api/v1/translate/batch); the remaining 9 are headline-counted additional distinct-scope confirmations on rows that were already Confirmed (EPUB→EPUB round-trip, -script Cyrillic→Latin, preparation analysis, GET /api/v1/version, preparation→translate two-stage) or language-pair/provider confirmations

covered by already-Confirmed generic rows (EN→KO, DE→EN, FR→ES, EN→SR-latin) — counted in the operator-sanctioned running total but honestly distinguished here so no row is double-flipped. Wave 6 of 2026-06-16 added **5 net-new** via the default in-process direct DeepSeek CLI path, each ffprobe+frame-verified per §11.4.107 (790×560 yuv420p, advancing frames, English input-frame ≠ translated output-frame): **HTML→HTML round-trip** EN→ES (helix_translate-html_html_roundtrip-wave6-20260616-175612.mp4, real Spanish HTML), **PDF input accepted** EN→ES (helix_translate-pdf_input-wave6-20260616-175612.mp4, from a real PDF 1.7), **DOCX input accepted** EN→ES (helix_translate-docx_input-wave6-20260616-175612.mp4, from a real Word 2007+ file), **EN→ZH (Chinese)** (helix_translate-en_zh_translate-wave6-20260616-175612.mp4) and **EN→RU (Russian)** (helix_translate-en_ru_translate-wave6-20260616-175612.mp4); Wave 6 also recorded an honest MINOR finding (HTML/EPUB chapter-title duplication on output, PENDING_FORENSICS, cosmetic — core translation correct). Wave 5 of 2026-06-16 added **4 net-new** (the -multipass row was NOT counted — see §11.4.138 demotion below): **EN→PT (Portuguese)** (helix_translate-en-pt-translate-wave5-20260616-173814.mp4), **EN→JA (Japanese, non-Latin)** (helix_translate-en-ja-translate-wave5-20260616-173814.mp4), **REST POST /api/v1/convert/script** (helix_translate-rest-convert-script-wave5-20260616-173814.mp4, real Cyrillic↔Latin JSON both directions) and **REST POST /api/v1/translate/fb2** (helix_translate-rest-translate-fb2-wave5-20260616-173814.mp4, FB2→real Serbian EPUB; note: handler hardcodes ru→sr — tracked fix). Wave 4 of 2026-06-16 added **3 net-new** multi-language-pair translations from the same EN source proving genuine per-language LLM calls (distinct outputs): **EN→FR** (helix_translate-multilang-en-fr-wave4-20260616-173316.mp4, “Le soleil matinal se leva...”), **EN→DE** (helix_translate-multilang-en-de-wave4-20260616-173319.mp4, “Der morgendliche Sonnenaufgang...”), **EN→IT** (helix_translate-multilang-en-it-wave4-20260616-173419.mp4, “Il sole mattutino sorse...”); Wave 4 also re-confirmed (NOT double-counted) **RU→SR-Cyrillic** (helix_translate-multilang-ru-sr-cyr-wave4-20260616-173445.mp4). **§11.4.138 DEMOTION (-1):** the pkg/verification **Multi-pass polisher** (-multipass) row, previously PASS via helixtranslate-cli-multipass-verify-20260615.mp4, is DEMOTED to PENDING_FORENSICS — the wave-3e regression (docs/qa/wave3e-multipass-regression/FINDING.md) and wave-5 evidence prove the default-model path (gpt-4) fails the ValidModels whitelist and silently no-ops, and even with an explicit model the polish path can write an LLM meta-response instead of polished prose; a tracked source fix + re-record is owed before any re-confirmation. Net video-confirmed: 43 (Rev 14) – 1 (multipass demote) + 3 (Wave 4) + 4 (Wave 5) + 5 (Wave 6) = **54**. Wave 3f of 2026-06-16 added **2 net-new** — **EPUB→EPUB conversion** (helix_translate-epub-output-wave3e-20260616.mp4: EN .epub → real deepseek run → OUTPUT EPUB carries real Spanish “La Antigua Farola la vieja farola se alzaba sola en la rocosa costa...”) and **TXT→TXT translation** (helix_translate-txt-translation-wave3f-20260616.mp4: EN .txt → real deepseek run → ES .txt “El río fluía tranquilamente a través del valle verde. Los pájaros cantaban bajo la luz de la mañana.”), each ffprobe+frame-verified per §11.4.107 (790×560 yuv420p,

≥ 6 frames, start-frame English-only $\neq$ output-frame Spanish); Wave 3f also **re-confirmed** (NOT double-counted) the already-confirmed **FB2 $\rightarrow$ FB2** (helix_translate-fb2-convert-wave3e-20260616.mp4, namespace preserved + real Spanish), **format-detection** (helix_translate-format-detect-wave3e-20260616.mp4, extensionless .dat detected-by-content), **-script Cyrillic $\rightarrow$ Latin** (helix_translate-script-conversion-wave3e-20260616.mp4, RUN A latin “Ovo je testni...” $\neq$ RUN B cyrillic “Систем треба...”) and **markdown-translator EPUB $\rightarrow$ MD round-trip** (helix_translate-markdown-roundtrip-wave3f-20260616.mp4, “# El Faro Viejo / El viejo faro se alzaba solitario en la costa rocosa.”) rows with fresh frame-verified clips; running total 41 (Rev 13) + 2 = **43**. Wave 3d of 2026-06-16 added **4 net-new**, prioritising feature-type diversity — the **cmd/verify-models CLI** (real LLMsVerifier discovery/scoring/selection: 14 env-keyed providers with real per-provider reachable/auth/HTTP-status/candidate/verified counts, summary total verified: 4, then model-bridge list selection ranking #1 novita/Sao10K/L3-8B-Stheno-v3.2 score=0.91875 ... #4 siliconflow score=0.88125 — helix_translate-verify-models-wave3d-20260616.mp4), the **cmd/workable-items CLI** (real §11.4.93 SQLite SSOT: validate $\rightarrow$ OK: DB ... matches markdown (81 items), list $\rightarrow$ real ATM-NNN rows + descriptions — helix_translate-workable-items-wave3d-20260616.mp4), and **2 more providers** — **nvidia** (“...Un anciano pescador **caminaba** a lo largo de la orilla.” via integrate.api.nvidia.com, helix_translate-provider-nvidia-translate-wave3d-20260616.mp4) and **openrouter** (“...Un anciano pescador **paseaba** a lo largo de la orilla.” via openrouter.ai/api, helix_translate-provider-openrouter-translate-wave3d-20260616.mp4), the two provider outputs distinct (caminaba/paseaba) proving genuine per-provider models, all four Wave-3d clips fprobe+frame-verified per §11.4.107 (790 $\times$ 560 yuv420p, ≥ 6 frames at fps=2, CLI clips show advancing distinct sections / provider clips show start-frame English-only $\neq$ output-frame Spanish); the Wave-3d providers reach the llm_provider/OpenAI-compatible per-provider path; upstage was tested but is an honest SKIP (auth=false HTTP 403), together SKIPped (no key) — never faked. Wave 3c of 2026-06-16 added **2 net-new** — the **Hyperbolic** library client (pkg/translator/llm) and the **Fireworks** provider in the llm_provider submodule (llm_provider/pkg/providers/fireworks/fireworks.go), each a real-use Wave-3c .mp4 showing a distinct EN $\rightarrow$ ES translation through the OpenAI-compatible per-provider path — helix_translate-provider-hyperbolic-translate-wave3c-20260616.mp4 (“Al amanecer, el sol salía sobre el tranquilo pueblo. Un anciano pescador caminaba por la orilla...”) and helix_translate-provider-fireworks-translate-wave3c-20260616.mp4 (“Al amanecer, el sol ascendió sobre el tranquilo pueblo. Un anciano pescador paseaba por la orilla...”), the two outputs distinct from each other (salía/caminaba vs ascendió/paseaba) proving genuine per-provider models, fprobe+frame-verified per §11.4.107 (790 $\times$ 560 yuv420p, ≥ 6 frames at fps=2, start-frame input-only $\neq$ output-frame Spanish). Wave 3c also **re-confirmed** (no double-count) the already-confirmed REST POST /api/v1/translate and GET /health · /api/v1/version · /api/v1/providers rows against a freshly-started live cmd/server TLS instance (port 9443) — helix_translate-rest-translate-wave3c-20260616.mp4 (real JSON {"translated":"El sol de la mañana se alzó sobre el tranquilo pueblo.", "provider":"llm-deepseek",...}) and

helix_translate-rest-endpoints-wave3c-20260616.mp4 (real health/version/providers JSON). The GET /api/v1/verified-models 404 remains an honest PENDING gap (LLMsVerifier disabled in config — not confirmed). Wave 3b of 2026-06-16 added 3 net-new — the **Gemini**, **Zhipu (GLM)** and **Cohere** library clients, each backed by a real-use Wave-3b .mp4 showing a distinct EN→SR-latin translation through the per-provider path — helix_translate-provider-gemini-translate-wave3b-20260616.mp4 (“Staro svjetiljko stajalo je samotočno na kamenitoj obali...”), helix_translate-provider-zhipu-translate-wave3b-20260616.mp4 (“Stara svjetiljka stojeća samostalno na kamenitoj obali...”), helix_translate-provider-cohere-translate-wave3b-20260616.mp4 (provider=cohere via -base-url https://api.cohere.ai/compatibility/v1, “Stari svjetionik stajao je sam na kamenitoj obali... svijeća njegova vodila brodove...”), all four outputs (incl. the re-confirmed deepseek clip helix_translate-provider-deepseek-translate-wave3b-20260616.mp4 “Stari svjetionik stajao je sam na kamenitoj obali...”) distinct from one another proving genuine per-provider models, fprobe-verified (non-zero dur+frames, 790×560 even-dims yuv420p) + ≥6 frames extracted with one content-verified per §11.4.107; the deepseek clip is a re-confirmation, no double-count; Wave 3a of 2026-06-16 added 2 net-new — the **Cerebras** and **SambaNova** per-provider LLM clients, each backed by a real-use Wave-3a .mp4 showing a distinct EN→ES translation through unified-translator -provider <name> — helix_translate-provider-cerebras-translate-wave3a-20260616.mp4 (“El sol de la mañana se alzó sobre el tranquilo pueblo.”) and helix_translate-provider-sambanova-translate-wave3a-20260616.mp4 (“El sol matutino se alzó sobre el tranquilo pueblo.”), the two outputs distinct from each other proving genuine per-provider models, fprobe+frame-verified per §11.4.107 with start-frame ≠ output-frame liveness; Wave 2 of 2026-06-16 added 5 net-new — DOCX output, PDF output, HTML→Markdown conversion, ProviderDiverseTranslators, EnsembleFactory — each backed by a real-use Wave-2 .mp4 independently validator-verified fprobe+frame; Wave 2 also re-confirmed the already-confirmed gRPC TranslationService row with an additional clip, no double-count; Wave 1 added 2 — bridge-ensemble best-model selection + bridge provider-diverse catalogue) across unified-translator (DeepSeek translate, EPUB→TXT / HTML→EPUB / FB2→EPUB / FB2→FB2 conversions, Serbian-Cyrillic, -script Cyrillic↔Latin, format-detection, help/version, -verify, **-multipass multi-pass polishing engine**), the REST API server (POST /api/v1/translate real DeepSeek + target_lang FIXED, TLS startup FIXED, /health · /api/v1/version · /api/v1/providers), the gRPC server (real EN→ES), markdown-translator (EPUB↔MD round-trip), preparation-translator (FIXED real analysis), cmd/cli (real DeepSeek), monitor-server (live WS hub), the DeepSeek library client, and the **LLMsVerifier→agent bridge CLI** (cmd/model-bridge — helixtranslate-bridge-bestmodel-translate-20260615.mp4: in-process verify→SQLite store→select strongest novita/Sao10K/L3-8B-Stheno-v3.2 score 0.919 → live invoke EN→ES “Good morning, friend.” → “Buenos días, amigo.”; fprobe 8.0s/80 frames, ≥6 frames content-verified) **and the LLMsVerifier→agent bridge MCP-stdio surface** (cmd/model-bridge mcp — helixtranslate-bridge-mcp-stdio-20260615.mp4: a real JSON-RPC MCP session over stdio — initialize→tools/list→bridge_best_model→bridge_invoke — where tool bridge_invoke returned a live verified-model translation “Le

pont relie deux rives.” (isError:None); ffprobe 11.88s/10 frames, ≥6 frames content-verified). **Now video-confirmed (Wave 2): DOCX output** (Implemented; pure-Go OOXML writer pkg/ebook/docx_writer.go, commit 87cd2be) — Wave-2 real-use video helix_translate-docx-output-20260616-112943.mp4 (unified-translator DOCX output, openai/gpt-4, the produced .docx carries real Spanish text; validator ffprobe+frame-verified). **PDF output** (Implemented, commit fb265e7; pkg/ebook/pdf_writer.go Book→HTML5→weasyprint→Cyrillic-faithful PDF, wired into unified-translator .pdf output, main.go:1160) — Wave-2 real-use video helix_translate-pdf-output-20260616-113057.mp4 (valid PDF with real Spanish; validator ffprobe+frame-verified). **Wave-1 bridge-translator-path blocker RESOLVED (caveat 8):** the ValidModels-whitelist runtime failure that blocked the DOCX/PDF/gRPC/per-provider videos in Wave 1 is fixed — the verified-bridge translator path now produces real per-provider translations (Wave-2 provider clips: novita/mistral/groq/siliconflow). **Honest gaps recorded, NOT confirmed:** cmd/translator + translate-ssh + ebook-translator are **Obsolete** (→ **Fixed.md**), **Rev 7** (feature-removed — the local-runtime + SSH-local paths were removed in bridge phase-2 R-2..R-4; the default path now sources the LLMsVerifier pkg/bridge; the earlier STUB/OPERATOR-BLOCKED wording is stale and corrected); GET /api/v1/verified-models returns 404 because LLMsVerifier is disabled in config. Everything else is **PENDING** (a recording is owed) or **N/A** (no user-visible video applies — e.g. CLI flags, internal library types). Absence of a runtime recording is stated honestly, never papered over.

Operator-blocked items (read first, §11.4.45)

Two binaries are OPERATOR-BLOCKED — both require a remote SSH worker host running llama.cpp, which is not available in this environment. The binaries build and run as far as the missing host allows (captured in helixtranslate-cmd-blocked-binaries_20260615-172456.mp4):

- **cmd/translate-ssh** — SSH worker 4-step FB2→EPUB translator. **Unblock condition:** a reachable SSH host with llama.cpp + a model.
- **cmd/ebook-translator** — FB2 remote-translate workflow (syncs binary to remote, runs remote translate, downloads outputs). **Unblock condition:** same remote SSH/llama.cpp host.
- **cmd/translator (remote path)** — remote SSH/llama.cpp translate path is OPERATOR-BLOCKED (same host requirement); its LOCAL path is a separate STUB (“local translation not yet implemented”).

Coverage summary

Metric	Value
Headline feature total (inventory’s deduplicated per-category tally)	417
Enumerated feature rows in this doc	497 (Rev 16 = 496 Rev 14 + 1 new cmd/grpc-translate-probe row — standalone gRPC translate-round-trip

Metric	Value
Implemented	probe client, Implemented + Wired + Source-confirmed, Video-confirmation PENDING; Rev 14 = 494 Rev 13 + 2 new unified-translator rows added Wave 3f for the net-new videos: EPUB→EPUB conversion + TXT→TXT translation; Rev 9 = 493 Rev 8 + 1 new unified-translator HTML→Markdown conversion row added that wave for the Wave-2 html-to-md video; §11.4.6 — re-derive with awk over ^\ 9-column rows. The earlier “478” was NOT re-derivable from the live table — it was a hand-tracked running tally that had drifted; Rev 8 added the 7 missing pkg/bridge ensemble-seam rows: BestTranslator, BestTranslatorFunc, EnsembleFactory, ProviderDiverseTranslators, ProviderDiverseClients, ProviderDiverseModels, BestClient) 446 (Rev 16: 445 Rev 14 + 1 new cmd/grpc-translate-probe row, Implemented + Wired; Rev 14: 443 Rev 13 + 2 new unified-translator rows EPUB→EPUB conversion + TXT→TXT translation, both Implemented + Wired + Wave-3f video-confirmed; Rev 9: 442 Rev 8 + 1 new unified-translator HTML→Markdown conversion row; Rev 8 was 435 Rev 7 + 7 new pkg/bridge ensemble-seam rows; was 471 Rev 6, -36 flipped to Obsolete Rev 7 — local-runtime + SSH-local features removed in bridge phase-2 R-2..R-4)
Obsolete (→ Fixed.md)	39 (Rev 7 — bridge phase-2 R-2..R-4 removals: cmd/translate-ssh ×9, cmd/ssh-translation ×2, cmd/translator ×19, cmd/ebook-translator ×3, pkg/sshworker ×2, pkg/modelsbridge ×1, Ollama provider ×1, llama.cpp provider+coordinator ×2; Reason=feature-removed; the KEPT distributed-API path is NOT Obsolete)
Stub	4 (POST /api/v1/translate/ebook, POST /api/v1/preparation/analyze, GET /api/v1/preparation/result/:id, POST /api/v1/translate-with-verification)
Not implemented (gap rows)	0 (PDF-write flipped gap→Implemented Rev 4, commit fb265e7; output now EPUB/FB2/TXT/HTML/MD/DOCX/PDF)

Metric	Value
Partial	2 (POST /api/batch on standalone pkg/api/server.go — returns queued batch_id only, no translation; vision_engine OpenCV — stub default, real impl behind build tag) — was 3 Rev 6; cmd/translator flipped to Obsolete Rev 7
Operator-blocked	0 (was 2 Rev 6 — translate-ssh + ebook-translator flipped to Obsolete Rev 7: the remote SSH/llama.cpp worker capability was removed, not merely host-blocked)
Video-confirmed	66 (real, content-verified recordings — see Anti-bluff note. Rev-16 / Wave 9 + Wave 8 + Wave 7 of 2026-06-16 added 12 net-new (4+5+3), of which 3 are genuine row-flips PENDING→Confirmed (POST /api/v1/translate/string Wave 7, the Groq client library row Wave 8, POST /api/v1/translate/batch Wave 9) and 9 are headline-counted additional distinct-scope confirmations on already-Confirmed rows / language-pair coverage (EPUB→EPUB round-trip, -script Cyrillic→Latin, preparation analysis, GET /api/v1/version, preparation→translate two-stage, EN→KO, DE→EN, FR→ES, EN→SR-latin — see Anti-bluff note for the full per-feature mapping + the Wave-7 markdown-translator bug found→fixed-at-4fbe581→re-record-owed + the Wave-9 honest non-confirmable gaps). §11.4.6 cell reconciliation: this cell had a known sync-lag behind the authoritative headline running total (it read 43 at Rev 15 while the Anti-bluff-note prose was at 54); it is corrected here to the true running total — 54 (prior authoritative) + 4 (Wave 7) + 5 (Wave 8) + 3 (Wave 9) = 66. Rev-14 / Wave 3f of 2026-06-16 added 2 net-new video confirmations prioritising feature-type diversity — EPUB→EPUB conversion (helix_translate-epub-output-wave3e-20260616.mp4, EN .epub → real Spanish EPUB) and TXT→TXT translation (helix_translate-txt-translation-wave3f-20260616.mp4, EN .txt → real Spanish .txt), each ffmpeg+frame-verified per §11.4.107; Wave 3f also re-confirmed (NOT double-counted) FB2→FB2 /

Metric	Value
	format-detection / -script Cyrillic[?]Latin / markdown EPUB[?]MD round-trip with fresh frame-verified clips; running total 41 (Rev 13) + 2 = 43. Rev-13 / Wave 3d of 2026-06-16 added 4 net-new video confirmations prioritising feature-type diversity — the cmd/verify-models CLI (helix_translate-verify-models-wave3d-20260616.mp4, real LLMsVerifier discovery/scoring/selection + ranking), the cmd/workable-items CLI (helix_translate-workable-items-wave3d-20260616.mp4, real §11.4.93 SQLite validate+list), and 2 more providers nvidia (helix_translate-provider-nvidia-translate-wave3d-20260616.mp4, “...caminaba a lo largo de la orilla.”) + openrouter (helix_translate-provider-openrouter-translate-wave3d-20260616.mp4, “...paseaba a lo largo de la orilla.”), each ffprobe+frame-verified per §11.4.107; running total 37 (Rev 12) + 4 = 41. upstage HTTP-403 SKIP + together no-key SKIP, never faked. Rev-12 / Wave 3c of 2026-06-16 added 2 net-new video confirmations — the Hyperbolic library client (helix_translate-provider-hyperbolic-translate-wave3c-20260616.mp4, “...el sol salía... caminaba por la orilla...”) and the Fireworks llm_provider submodule client (helix_translate-provider-fireworks-translate-wave3c-20260616.mp4, “...el sol ascendió...paseaba por la orilla...”), each a distinct real EN→ES translation via the OpenAI-compatible per-provider path, ffprobe+frame-verified per §11.4.107 (790×560 yuv420p, ≥6 frames at fps=2, start-frame input-only ≠ output-frame Spanish), the two outputs distinct proving genuine per-provider models; Wave 3c also re-confirmed (NOT double-counted) the already-confirmed REST POST /api/v1/translate + /health · /api/v1/version · /api/v1/providers rows against a live cmd/server TLS instance (helix_translate-rest-translate-wave3c-20260616.mp4 + helix_translate-rest-endpoints-

Metric	Value
	wave3c-20260616.mp4); count reconciliation (§11.4.6): this cell read 32 through Rev 10/11 — a sync lag behind the Anti-bluff-note prose; it is corrected to the true running total here — 30 (Rev 9) + 2 (Wave 3a Cerebras/SambaNova) + 3 (Wave 3b Gemini/Zhipu/Cohere) + 2 (Wave 3c Hyperbolic/Fireworks) = 37. Rev-10 / Wave 3a of 2026-06-16 added 2 net-new video confirmations — the Cerebras and SambaNova per-provider LLM clients (helix_translate-provider-cerebras-translate-wave3a-20260616.mp4 + helix_translate-provider-sambanova-translate-wave3a-20260616.mp4), each a distinct real EN→ES translation via unified-translator -provider <name>, ffmpeg+frame-verified per §11.4.107 with start-frame ≠ output-frame liveness, the two outputs distinct proving genuine per-provider models; Rev-11 / Wave 3b of 2026-06-16 added 3 net-new — the Gemini, Zhipu (GLM) and Cohere library clients (helix_translate-provider-{gemini,zhipu,cohere}-translate-wave3b-20260616.mp4), distinct EN→SR-latin per-provider translations, frame-verified per §11.4.107; Rev-9 / Wave 2 of 2026-06-16 added 5 net-new video confirmations, each backed by a real-use Wave-2 .mp4 independently validator-verified ffmpeg+frame §11.4.107: DOCX output (helix_translate-docx-output-20260616-112943.mp4), PDF output (helix_translate-pdf-output-20260616-113057.mp4), HTML→Markdown conversion (helix_translate-html-to-md-20260616-114228.mp4), ProviderDiverseTranslators + EnsembleFactory (the 4 per-provider clips novita/mistral/groq/siliconflow — helix_translate-provider-{novita,mistral,groq,siliconflow}-translate-20260616-114*.mp4). Wave 2 also re-confirmed the already-confirmed gRPC TranslationService row with an additional clip helix_translate-grpc-translate-20260616-113415.mp4 (NOT double-counted — that row was

Metric	Value
	already video-confirmed). DOCX output and PDF output are no longer “real-artifact-only” — they are now video-confirmed. §11.4.108 note: the Wave-1 bridge-translator-path runtime blocker (caveat 8) is RESOLVED, which is what made the DOCX/PDF/gRPC/per-provider Wave-2 recordings genuinely demonstrate working features (recording them in Wave 1 would have been a §11.4.107 bluff; they were honest SKIPS then). Wave 1 added 2: bridge-ensemble best-model selection + bridge provider-diverse catalogue. $\approx 8.3\%$ of 494 enumerated rows ($\approx 9.8\%$ of the 417 headline)
Video-confirmation PENDING	runtime/user-visible features without a recording yet
Video-confirmation N/A	flags, internal library types, infra middleware (no standalone user-visible video)

§11.4.6 reconciliation: The inventory’s headline is **417** — its own note states this is an *approximate, deduplicated* per-category tally where “some rows are sub-features... counted once, and a few rows span two categories (counted once, in the first/primary category).” The inventory’s *detailed* tables actually enumerate **470** feature rows. This document transcribes **all 470 detailed rows 1:1** so **no feature is dropped** (which would itself be a bluff). The 417 headline and the 470 enumerated rows are both reported honestly; the 53-row gap is dedup, not missing coverage.

“Static-return” endpoints (e.g. GET /api/stats hardcoded zeros, GET /api/v1/status/:id hardcoded completed, GET /api/v1/metrics static, monitor-server GET /api/v1/status static monitoring_active) are marked **Implemented** at the route level but their Real-use cell states the static/hardcoded behaviour honestly so no false “working backend” impression is created.

Per-category counts

Category	Count
CLI	130
API	70
Web	23
gRPC	7
Library	81
Submodule	95
Infra	11
TOTAL	417

Legend

- **Implementation:** Implemented (real code path present) / Stub (returns mock, no real work) / Partial (does some but not the named work).
- **Wiring:** Wired (reachable from a binary / route / public API) / Unwired (present but not reached).
- **Real-use:** what the code actually does at runtime (honest, §11.4.6).
- **Tests:** Covered (unit/integration/challenge present in repo) / None (no test found / not asserted in inventory).
- **Validation:** strongest evidence available.
- **Video-confirmation:** Confirmed:<file> only with a real recording; else PENDING / N/A.

Per §11.4.6, where the inventory did not read test files line-by-line, Tests is recorded Not-inventoried (honest unknown) rather than guessed as Covered or None. The repo is known to carry extensive `_test.go` suites and a `test/` tree, but per-feature test attribution was not part of the raw inventory.

1. CLI Binaries — cmd/

cmd/api-server

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	V
api-server	REST translation server	API/Web	Implemented	Wired	Gin HTTP server (:8080) proxying to gRPC backend; serves dashboard + static files	Not-inventoried	S co (c se m
api-server	-http-port flag	API	Implemented	Wired	Override HTTP port (else HTTP_PORT env)	Not-inventoried	S co
api-server	-http-address flag	API	Implemented	Wired	Override HTTP bind address (else HTTP_ADDRESS env)	Not-inventoried	S co
api-server		API	Implemented	Wired	Override gRPC backend	Not-inventoried	S co

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	V
	-grpc-address flag				address (else GRPC_ADDRESS env)		
api-server	-debug flag	API	Implemented	Wired	Enable debug / Gin debug mode	Not-inventoried	S co

cmd/challenge-runner

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Va
challenge-runner	Challenge orchestrator	CLI	Implemented	Wired	Registers + runs all anti-bluff challenges, prints PASS/ FAIL/ SKIP, writes results dir	Not-inventoried	Sc co
challenge-runner	-parallel	CLI	Implemented	Wired	Run challenges in parallel	Not-inventoried	Sc co
challenge-runner	-max-concurrency	CLI	Implemented	Wired	Max concurrent challenges	Not-inventoried	Sc co
challenge-runner	-stop-on-failure	CLI	Implemented	Wired	Stop on first failure	Not-inventoried	Sc co
challenge-runner	-verbose	CLI	Implemented	Wired	Verbose output	Not-inventoried	Sc co
challenge-runner	-timeout	CLI	Implemented	Wired	Per-challenge timeout	Not-inventoried	Sc co
challenge-runner	-filter	CLI	Implemented	Wired	Comma-separated challenge IDs to run	Not-inventoried	Sc co
challenge-runner	-category	CLI	Implemented	Wired	Limit to a category	Not-inventoried	Sc co
challenge-runner	-results-dir	CLI	Implemented	Wired	Results output directory	Not-inventoried	Sc co

cmd/cli

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Ver
cli	Universal ebook translator	CLI	Implemented	Wired	Parse ebook → translate → atomic-write EPUB/FB2/TXT; multi-llm/distributed support	Not-inventoried	Parse ebook → translate → atomic-write EPUB/FB2/TXT; multi-llm/distributed support (default: sa, W (8) va w sil os su pa G no (5) pr
cli	-i/-input	CLI	Implemented	Wired	Input ebook (FB2/EPUB/TXT/HTML/PDF/DOCX)	Not-inventoried	Sc
cli	-o/-output	CLI	Implemented	Wired	Output file	Not-inventoried	Sc
cli	-f/-format	CLI	Implemented	Wired	Output format (epub, fb2, txt)	Not-inventoried	Sc
cli	-p/-provider	CLI	Implemented	Wired	Translation provider	Not-inventoried	Sc
cli	-model	CLI	Implemented	Wired	LLM model name	Not-inventoried	Sc
cli	-api-key	CLI	Implemented	Wired	Provider API key	Not-inventoried	Sc
cli	-base-url	CLI	Implemented	Wired	Provider base URL	Not-inventoried	Sc
cli	-script	CLI	Implemented	Wired	Output script (default/cyrillic/latin)	Not-inventoried	Sc
cli	-locale	CLI	Implemented	Wired			Sc

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Ver
					Target language locale code	Not-inventoried	
cli	-language	CLI	Implemented	Wired	Target language name	Not-inventoried	Sc
cli	-source	CLI	Implemented	Wired	Source language (auto-detect if unset)	Not-inventoried	Sc
cli	-detect	CLI	Implemented	Wired	Detect source language and exit (LLM detector if API key)	Not-inventoried	Sc
cli	-v/-version	CLI	Implemented	Wired	Show version	Not-inventoried	Sc
cli	-h/-help	CLI	Implemented	Wired	Show help	Not-inventoried	Sc
cli	-create-config	CLI	Implemented	Wired	Write a config-file template	Not-inventoried	Sc
cli	-disable-local-llms	CLI	Implemented	Wired	Use only distributed/API providers	Not-inventoried	Sc
cli	-prefer-distributed	CLI	Implemented	Wired	Prefer distributed workers	Not-inventoried	Sc
cli	-c/-config	CLI	Implemented	Wired	Config file path	Not-inventoried	Sc
cli	-hash-codebase	CLI	Implemented	Wired	Compute codebase hash and exit	Not-inventoried	Sc

cmd/deployment

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
deployment	Distributed deployment manager	CLI	Implemented	Wired	Orchestrates Docker-based main+worker deployment over SSH	Not-invent

Not-inventoried | Source-confirmed | N/A | | ebook-translator | Output verification | CLI | **Obsolete (→ Fixed.md)** | Wired | Verifies MD/EPUB exist, non-degenerate, target-language (Cyrillic) check | Not-inventoried | Source-confirmed | PENDING |

cmd/grpc-server

Component	Feature	Category	Implementation	Wiring	Real-use	
grpc-server	gRPC TranslationService	gRPC	Implemented	Wired	gRPC server (:50051): Start/Status/List/Cancel/Stream/GetProviders/SubscribeEvents	M i
grpc-server	-address	gRPC	Implemented	Wired	Bind address (or GRPC_ADDRESS env)	M i
grpc-server	-port	gRPC	Implemented	Wired	Port (or GRPC_PORT env)	M i
grpc-server	-max-connections	gRPC	Implemented	Wired	Max concurrent translations	M i
grpc-server	-reflection	gRPC	Implemented	Wired	Enable gRPC reflection (or ENABLE_REFLECTION env)	M i
grpc-server	-metrics	gRPC	Implemented	Wired	Enable metrics (or ENABLE_METRICS env)	M i
grpc-server	-log-level	gRPC	Implemented	Wired	Log level (or LOG_LEVEL env)	M i
grpc-server	-version	gRPC	Implemented	Wired	Show version	M i
grpc-server	-help	gRPC	Implemented	Wired	Show help	M i

cmd/grpc-translate-probe

Component	Feature	Category	Implementation	Wiring	Real-use
grpc-translate-probe	Standalone gRPC translate-round-trip probe client	gRPC	Implemented	Wired	Pure wire-protocol client (cmd/grpc-translate-probe). TranslationService (default name) StartTranslation(input_file) then polls GetTranslationStatus() until COMPLETED/FILE lines + generated file must pre-exist on the server fs (see flags -addr/-input/-output/-session/-timeout; exit 0 complete)

cmd/markdown-translator

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
markdown-translator	EPUB → Markdown translate pipeline	CLI	Implemented	Wired	EPUB/MD → markdown → translate → EPUB/MD; optional preparation phase	Not-invented
markdown-translator	-input	CLI	Implemented	Wired	Input file (EPUB or Markdown)	Not-invented
markdown-translator	-output	CLI	Implemented	Wired	Output file (auto if empty)	Not-invented
markdown-translator	-format	CLI	Implemented	Wired	Output format (epub, md)	Not-invented

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
markdown-translator	-lang	CLI	Implemented	Wired	Target language code	Not-inventoried
markdown-translator	-provider	CLI	Implemented	Wired	LLM provider (deepseek/openai/anthropic/llamacpp)	Not-inventoried
markdown-translator	-model	CLI	Implemented	Wired	LLM model	Not-inventoried
markdown-translator	-keep-md	CLI	Implemented	Wired	Keep intermediate markdown	Not-inventoried
markdown-translator	-prepare	CLI	Implemented	Wired	Enable multi-LLM preparation/analysis phase	Not-inventoried
markdown-translator	-prep-passes	CLI	Implemented	Wired	Number of preparation passes	Not-inventoried

cmd/monitor-server

Component	Feature	Category	Implementation	Wiring	Real-use
monitor-server	WebSocket monitoring hub	Web	Implemented	Wired	Gin server (:8090, MONITOR_SERVER_PORT) /ws, /monitor dashboard status + health
monitor-server	GET /ws	Web	Implemented	Wired	WebSocket subscription by session_id
monitor-server	/monitor + /	Web	Implemented	Wired	Serve monitor.html (redirect from /)
monitor-server	GET /api/v1/status/:session_id	Web/API	Implemented	Wired	Returns static monitoring_action JSON (not a live session lookup)
monitor-server	GET /health	Web	Implemented	Wired	Health + client count
monitor-server	MONITOR_SERVER_PORT env	Web	Implemented	Wired	Configure bind port (CLI flag)

cmd/preparation-translator

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
preparation-translator	Preparation+translation runner	CLI	Implemented	Wired	Parse ebook → multi-LLM preparation analysis → translate → write EPUB + analysis JSON	Not-invent
preparation-translator	-input	CLI	Implemented	Wired	Input ebook path	Not-invent
preparation-translator	-output	CLI	Implemented	Wired	Output EPUB path	Not-invent
preparation-translator	-analysis	CLI	Implemented	Wired	Preparation analysis JSON output path	Not-invent
preparation-translator	-source	CLI	Implemented	Wired	Source language	Not-invent
preparation-translator	-target	CLI	Implemented	Wired	Target language	Not-invent
preparation-translator	-passes	CLI	Implemented	Wired	Number of preparation passes	Not-invent
preparation-translator	-providers	CLI	Implemented	Wired	Comma-separated LLM providers	Not-invent

cmd/server

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
server	Main TLS HTTP/2+HTTP/3 server	API/Web	Implemented	Wired	TLS REST/WebSocket API (config.json); CORS + rate	Not-inventoried

Source-confirmed | PENDING | | ssh-translation | (no CLI flags) | CLI | **Obsolete** (→ **Fixed.md**) | Wired | All config hardcoded in SystemConfig (host, creds, langs, llama providers) | Not-inventoried | Source-confirmed | N/A |

cmd/translate-ssh

Obsolete-Details: Since=bridge phase-2 R-2/R-4 (2026-06-16); Reason=feature-removed (local-runtime + SSH-local translation removed; default path now sources the LLMsVerifier bridge — no local llama.cpp/Ollama, no SSH-local worker); Superseding-item=pkg/bridge (LLMsVerifier→agent bridge); Operator-approval=2026-06-15 D1/D2 (forbid all local runtimes) + R-1 redirect + the R-4 “keep distributed API, remove SSH-local” decision (2026-06-16); Triple-check evidence=source files removed (FACT: git ls-files empty for the removed cmd/* dirs + pkg/sshworker + pkg/modelsbridge + ollama.go/llamacpp*.go) + CM-N0-LOCAL-RUNTIME gate PASS. The KEPT distributed-API path is NOT Obsolete. | Component | Feature | Category | Implementation | Wiring | Real-use | Tests | Validation | Video-confirmation | ||||| translate-ssh | SSH worker 4-step FB2→EPUB translator | CLI | **Obsolete** (→ **Fixed.md**) | Wired | FB2→MD → translate (remote llama.cpp) → MD→EPUB; standalone worker invoked by coordinator | Not-inventoried | **OPERATOR-BLOCKED** — needs a remote SSH worker host running llama.cpp; none available. Binary builds + runs as far as possible (blocked-binaries demo) | OPERATOR-BLOCKED; evidence:/Volumes/T7/Downloads/Recordings/helixtranslate-cmd-blocked-binaries_20260615-172456.mp4 || translate-ssh | -input | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Input ebook (required) | Not-inventoried | Source-confirmed | N/A || translate-ssh | -output | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Output EPUB (required) | Not-inventoried | Source-confirmed | N/A || translate-ssh | -host | CLI | **Obsolete** (→ **Fixed.md**) | Wired | SSH host (required) | Not-inventoried | Source-confirmed | N/A || translate-ssh | -user | CLI | **Obsolete** (→ **Fixed.md**) | Wired | SSH username (required) | Not-inventoried | Source-confirmed | N/A || translate-ssh | -password | CLI | **Obsolete** (→ **Fixed.md**) | Wired | SSH password (required) | Not-inventoried | Source-confirmed | N/A || translate-ssh | -port | CLI | **Obsolete** (→ **Fixed.md**) | Wired | SSH port | Not-inventoried | Source-confirmed | N/A || translate-ssh | -remote-dir | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Remote working directory | Not-inventoried | Source-confirmed | N/A || translate-ssh | -report-dir | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Report output directory | Not-inventoried | Source-confirmed | N/A |

cmd/translator

Obsolete-Details: Since=bridge phase-2 R-2/R-4 (2026-06-16); Reason=feature-removed (local-runtime + SSH-local translation removed; default path now sources the LLMsVerifier bridge — no local llama.cpp/Ollama, no SSH-local worker); Superseding-item=pkg/bridge (LLMsVerifier→agent bridge); Operator-approval=2026-06-15 D1/D2 (forbid all local runtimes) + R-1 redirect + the R-4 “keep distributed API, remove SSH-local” decision (2026-06-16); Triple-check evidence=source

files removed (FACT: git ls-files empty for the removed cmd/* dirs + pkg/sshworker + pkg/modelsbridge + ollama.go/llamacpp*.go) + CM-NO-LOCAL-RUNTIME gate PASS. The KEPT distributed-API path is NOT Obsolete. | Component | Feature | Category | Implementation | Wiring | Real-use | Tests | Validation | Video-confirmation | | | | | | | | | | translator | Documented FB2 local/remote translator | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Remote SSH/llama.cpp path produces a documentation report; LOCAL path is a STUB — emits “local translation not yet implemented” (captured). Remote path is OPERATOR-BLOCKED (no SSH worker host) | Not-inventoried | **GAP — local STUB captured; remote OPERATOR-BLOCKED** | PENDING/OPERATOR-BLOCKED; evidence:/Volumes/T7/Downloads/Recordings/helixtranslate-cmd-blocked-binaries_20260615-172456.mp4 | | translator | -i/-input | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Input ebook | Not-inventoried | Source-confirmed | N/A | | translator | -o/-output | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Output file | Not-inventoried | Source-confirmed | N/A | | translator | -ssh-host | CLI | **Obsolete** (→ **Fixed.md**) | Wired | SSH host (remote mode) | Not-inventoried | Source-confirmed | N/A | | translator | -ssh-user | CLI | **Obsolete** (→ **Fixed.md**) | Wired | SSH username | Not-inventoried | Source-confirmed | N/A | | translator | -ssh-password | CLI | **Obsolete** (→ **Fixed.md**) | Wired | SSH password | Not-inventoried | Source-confirmed | N/A | | translator | -ssh-port | CLI | **Obsolete** (→ **Fixed.md**) | Wired | SSH port | Not-inventoried | Source-confirmed | N/A | | translator | -remote-dir | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Remote working directory | Not-inventoried | Source-confirmed | N/A | | translator | -workers | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Parallel workers | Not-inventoried | Source-confirmed | N/A | | translator | -chunk-size | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Text chunk size | Not-inventoried | Source-confirmed | N/A | | translator | -concurrency | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Max concurrent operations | Not-inventoried | Source-confirmed | N/A | | translator | -verify | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Verify translated output | Not-inventoried | Source-confirmed | N/A | | translator | -verbose | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Verbose logging | Not-inventoried | Source-confirmed | N/A | | translator | -llama-binary | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Path to llama.cpp binary | Not-inventoried | Source-confirmed | N/A | | translator | -temperature | CLI | **Obsolete** (→ **Fixed.md**) | Wired | LLM temperature | Not-inventoried | Source-confirmed | N/A | | translator | -context | CLI | **Obsolete** (→ **Fixed.md**) | Wired | LLM context size | Not-inventoried | Source-confirmed | N/A | | translator | -version | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Show version | Not-inventoried | Source-confirmed | N/A | | translator | -help | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Show help | Not-inventoried | Source-confirmed | N/A | | translator | -hash-codebase | CLI | **Obsolete** (→ **Fixed.md**) | Wired | Compute codebase hash | Not-inventoried | Source-confirmed | N/A |

cmd/unified-translator (PRIMARY CLI)

Component	Feature	Category	Implementation	Wiring	Real-use	Test
unified-translator	Primary multi-provider translator	CLI	Implemented	Wired	Parse → MD → translate (API/llamacpp/ssh) → script-	Not-inve

Component	Feature	Category	Implementation	Wiring	Real-use	Test
unified-translator	DeepSeek translate (provider=deepseek)	CLI	Implemented	Wired	normalize → output (epub/fb2/html/txt/md) + session report EN→Serbian DeepSeek run, real LLM output written to TXT	Not-inve
unified-translator	EPUB→TXT conversion	CLI	Implemented	Wired	EPUB parse → DeepSeek translate → TXT writer, real Serbian text	Not-inve
unified-translator	HTML→EPUB conversion	CLI	Implemented	Wired	HTML parse → DeepSeek translate → EPUB writer	Not-inve
unified-translator	HTML→Markdown conversion	CLI	Implemented	Wired	HTML parse → translate (openai/gpt-4 via verified bridge) → Markdown (.md) writer; real Spanish md output	Not-inve
unified-translator	FB2→EPUB conversion	CLI	Implemented	Wired	FB2 parse → DeepSeek translate → EPUB writer	Not-inve
unified-translator	FB2→FB2 conversion	CLI	Implemented	Wired	FB2 parse → DeepSeek translate → FB2 writer (namespaces preserved)	Not-inve
unified-translator	Serbian Cyrillic output	CLI	Implemented	Wired	Translate with Serbian-Cyrillic target output	Not-inve
unified-translator	EPUB→EPUB conversion	CLI	Implemented	Wired	EPUB parse → DeepSeek translate →	Not-inve

Component	Feature	Category	Implementation	Wiring	Real-use	Test
					EPUB writer (round-trip, real translated text in produced .epub)	
unified-translator	TXT→TXT translation	CLI	Implemented	Wired	Plain-text parse → DeepSeek translate → TXT writer (real translated text, no format conversion)	Not-inve
unified-translator	HTML→HTML round-trip	CLI	Implemented	Wired	HTML parse → DeepSeek translate → HTML writer (round-trip, real translated text in produced .html)	Not-inve

Component	Feature	Category	Implementation	Wiring	Real-use	Test
unified-translator	PDF input accepted	CLI	Implemented	Wired	PDF parse → DeepSeek translate (real PDF 1.7 input)	Not-inve
unified-translator	DOCX input accepted	CLI	Implemented	Wired	DOCX parse → DeepSeek translate (real Word 2007+ input)	Not-inve
unified-translator	Format detection	CLI	Implemented	Wired	Input-format auto-detection on real input	Not-inve
unified-translator	help/version output	CLI	Implemented	Wired	-help/-version usage banner	Not-inve
unified-translator	-i/-input	CLI	Implemented	Wired	Input ebook	Not-inve
unified-translator	-o/-output	CLI	Implemented	Wired	Output file (auto if empty)	Not-inve
unified-translator	-source-lang / -target-lang (multi-language-pair capability)	CLI	Implemented	Wired	Source/target language selection driving real per-language LLM translation	Not-inve

Component	Feature	Category	Implementation	Wiring	Real-use	Test
unified-translator	-script (Serbian Cyrillic↔Latin)	CLI	Implemented	Wired	Target script (cyrillic/latin) script normalization on real run	Not-inve
unified-translator	-provider	CLI	Implemented	Wired	Provider (openai/anthropic/zhipu/deepseek/qwen/gemini/ollama/llamacpp/ssh)	Not-inve
unified-translator	-model	CLI	Implemented	Wired	Model name	Not-inve
unified-translator	-api-key	CLI	Implemented	Wired	Provider API key	Not-inve
unified-translator	-base-url	CLI	Implemented	Wired	Provider base URL	Not-inve
unified-translator	-temperature	CLI	Implemented	Wired	LLM temperature	Not-inve
unified-translator	-max-tokens	CLI	Implemented	Wired	Max tokens	Not-inve
unified-translator	-timeout	CLI	Implemented	Wired	Request timeout	Not-inve
unified-translator	-ssh-host	CLI	Implemented	Wired	SSH host (provider=ssh)	Not-inve
unified-translator	-ssh-user	CLI	Implemented	Wired	SSH username	Not-inve
unified-translator	-ssh-password	CLI	Implemented	Wired	SSH password	Not-inve

Component	Feature	Category	Implementation	Wiring	Real-use	Test
unified-translator	-ssh-port	CLI	Implemented	Wired	SSH port	Not-inve
unified-translator	-remote-dir	CLI	Implemented	Wired	Remote directory	Not-inve
unified-translator	-llama-binary	CLI	Implemented	Wired	Path to llama.cpp binary	Not-inve
unified-translator	-llama-model	CLI	Implemented	Wired	Path to llama.cpp model	Not-inve
unified-translator	-context-size	CLI	Implemented	Wired	LLM context size	Not-inve
unified-translator	-workers	CLI	Implemented	Wired	Parallel workers	Not-inve
unified-translator	-chunk-size	CLI	Implemented	Wired	Text chunk size	Not-inve
unified-translator	-concurrency	CLI	Implemented	Wired	Concurrent operations	Not-inve
unified-translator	-verify (output-quality check)	CLI	Implemented	Wired	Built-in per-step verification: input/markdown/translation-quality/valid-EPUB verdicts in session report. NOTE: this is the unified-translator's own verify pass, NOT the pkg/verification multipass engine (that engine is not wired into this CLI — see gap row below)	Not-inve
unified-translator	-verbose	CLI	Implemented	Wired	Verbose logging	Not-inve
unified-translator	-monitoring	CLI/Web	Implemented	Wired	Enable web monitoring	Not-inve
unified-translator	-monitoring-port	CLI/Web	Implemented	Wired	Monitoring server port	Not-inve
unified-translator	-use-verifier	CLI	Implemented	Wired	Use LLMsVerifier	Not-inve

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
unified-translator	-verifier-url	CLI	Implemented	Wired	for model selection LLMsVerifier API URL	Not-invented
unified-translator	-verifier-api-key	CLI	Implemented	Wired	LLMsVerifier API key	Not-invented
unified-translator	-version	CLI	Implemented	Wired	Show version	Not-invented
unified-translator	-help	CLI	Implemented	Wired	Show help	Not-invented

cmd/verify-models

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Verdict
verify-models	In-process LLMsVerifier runner	CLI	Implemented	Wired	Runs verification pipeline against env-keyed providers; persists verified models to SQLite	Not-inventoried	Partial LLMsVerifier scenarios can be verified through -model pipeline key providers re- state correct re- authentication can be optional state authenticated state signature verification supported verification performed through library sequence not State signature signature 09 signature W

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Ver
verify-models	-db	CLI	Implemented	Wired	SQLite store path	Not-inventoried	as ffj yu fp ve So
verify-models	-max	CLI	Implemented	Wired	Max models/provider through pipeline	Not-inventoried	So
verify-models	-min-score	CLI	Implemented	Wired	Min overall score to persist	Not-inventoried	So
verify-models	-timeout	CLI	Implemented	Wired	Overall verification timeout	Not-inventoried	So

cmd/model-bridge

LLMsVerifier→component+agent bridge (no local models): selects the best LLMsVerifier-scored model, exposes it to agents/components via a CLI and an MCP stdio server wired in .mcp.json. Source: cmd/model-bridge/{main.go,mcp.go}, pkg/bridge/, commit ab1bed3 (+ routing seam a5860b2).

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
model-bridge	LLMsVerifier→agent bridge CLI	CLI	Implemented	Wired	Selects best LLMsVerifier-scored model (top-1 + fallback chain) and translates/invokes through it; no local models	Cover bridg unit+ go t green sessi
model-bridge	MCP stdio server (.mcp.json)	CLI/Infra	Implemented	Wired	Exposes the selected model to	

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					MCP clients over stdio; wired as model-bridge in .mcp.json (./build/model-bridge)	Coverage mode mcp_

cmd/workable-items

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
workable-items	§11.4.93 workable-items SQLite tool	CLI	Implemented	Wired	Bidirectional sync between Issues/Fixed markdown and SQLite single-source-of-truth	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
workable-items	sync md-to-db	CLI	Implemented	Wired	Parse ATM-NNN headings from markdown into DB	Not-inventoried
workable-items	sync db-to-md	CLI	Implemented	Wired	Regenerate Issues_Summary/Fixed_Summary from DB	Not-inventoried
workable-items	validate	CLI	Implemented	Wired	Verify DB matches markdown; exit non-zero on drift	Not-inventoried
workable-items	diff	CLI	Implemented	Wired	Print md-vs-db drift human-readably	Not-inventoried
workable-items	list	CLI	Implemented	Wired	Print items in DB	Not-inventoried
workable-items	record-event	CLI	Implemented	Wired	Append event to item_history audit trail	Not-inventoried
workable-items	path/ value flags	CLI	Implemented	Wired	-issues/-fixed/-db/-issues-summary/-fixed-summary/-atm/-event/-on/-not	Not-inventoried

2. Library Packages — pkg/

pkg/translator/llm — LLM provider clients & factory

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
LLM Core	LLMClient interface & factory	Library	Implemented	Wired	LLMClient interface + NewLLMTranslator factory dispatching to ~32 providers	Not-inventoried
LLM Core	Type-coercion utilities	Library	Implemented	Wired	JSON config conversion helpers	Not-inventoried
LLM Core		Library	Implemented	Wired		

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
LLM Core	Verified factory	Library	Implemented	Wired	LLMsVerifier-gated translator construction w/ fallback chain (CONST-034)	Not-inventoried
	Mock client				Test mock LLMClient with fault injection (test-only)	Not-inventoried
	OpenAI client				GPT chat-completions translation	Not-inventoried
	Anthropic Claude client				Claude /v1/ messages translation	Not-inventoried
Google	Gemini client	Library	Implemented	Wired	Google Gemini generativelanguage API	Not-inventoried
DeepSeek	DeepSeek client	Library	Implemented	Wired	DeepSeek OpenAI-compatible chat-completions	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
Zhipu	Zhipu (GLM) client	Library	Implemented	Wired	Zhipu GLM /api/paas/v4	Not-inventori
Alibaba	Qwen client	Library	Implemented	Wired	Qwen/DashScope (OAuth2-capable)	Not-inventori
Ollama	Ollama local client	Library	Obsolete (→ Fixed.md)	Wired	Local /api/generate inference — REMOVED (bridge phase-2 R-2). See Obsolete-Details below.	Not-inventori
llama.cpp	llama.cpp client	Library	Obsolete (→ Fixed.md)	Wired	llama-cli wrapper w/ hardware detection, model download, GPU accel — REMOVED (bridge phase-2 R-3). See Obsolete-Details below.	Not-inventori
llama.cpp	llama.cpp multi-worker coordinator	Library	Obsolete (→ Fixed.md)	Wired	Worker pool for parallel local llama.cpp inference — REMOVED (bridge phase-2 R-3). See Obsolete-Details below.	Not-inventori

Obsolete-Details (Ollama + llama.cpp provider/coordinator rows above):
Since=bridge phase-2 R-2/R-3 (2026-06-16); Reason=feature-removed (local-runtime inference removed; the default path now sources the

LLMsVerifier bridge — no local llama.cpp/Ollama); Superseding-item=pkg/bridge; Operator-approval=2026-06-15 D1/D2 (forbid all local runtimes) + R-1 redirect; Triple-check evidence=pkg/translator/llm/{ollama.go, llamacpp.go, llamacpp_provider.go} removed (FACT: git ls-files empty) + CM-N0-LOCAL-RUNTIME gate PASS. The KEPT API provider clients (OpenAI/Anthropic/DeepSeek/...) in this section are NOT Obsolete. | Groq | Groq client | Library | Implemented | Wired | OpenAI-compatible wrapper | Not-inventoried | **PASS — real EN→ES translation captured** (Wave 8: build/unified-translator -provider groq -model llama-3.3-70b-versatile, real Groq API path: “The old lighthouse stood alone on the rocky cliff...” → “La vieja farola se erguía sola sobre el acantilado rocoso. Cada noche su haz barría el mar oscuro, guiando a las naves perdidas de vuelta a puerto.”; ffmpeg 115 frames/3.83s, source-frame ≠ output-frame per §11.4.107) | Confirmed:/Volumes/T7/Downloads/Recordings/helix_translate-groq-en-es-wave8-20260616_182021.mp4 | | Cohere | Cohere client | Library | Implemented | Wired | command-r via OpenAI-compat endpoint | Not-inventoried | **PASS — real Cohere translation captured** (Wave-3b real-use video: unified-translator -provider cohere -model command-r-08-2024 -base-url https://api.cohere.ai/compatibility/v1, real key, EN→SR-latin “Stari svjetionik stoeo je sam na kamenitoj obali. Noću svaki put svijeća njegova vodila brodove sigurno kući kroz oluju.”; log shows provider=cohere; ffmpeg 6.24s / 790×560 / non-zero frames; ≥6 frames extracted, f_04 content-verified per §11.4.107; output distinct from deepseek/gemini/zhipeu) | Confirmed:/Volumes/T7/Downloads/Recordings/helix_translate-provider-cohere-translate-wave3b-20260616.mp4 | | Mistral | Mistral client | Library | Implemented | Wired | OpenAI-compatible wrapper | Not-inventoried | Source-confirmed | PENDING | | xAI | xAI (Grok) client | Library | Implemented | Wired | OpenAI-compatible wrapper | Not-inventoried | Source-confirmed | PENDING | | Replicate | Replicate client | Library | Implemented | Wired | Async prediction-polling API | Not-inventoried | Source-confirmed | PENDING | | Cerebras | Cerebras client | Library | Implemented | Wired | OpenAI-compatible wrapper (https://api.cerebras.ai/v1, model llama3.1-70b) | Not-inventoried | **PASS — real EN→ES translation captured** (via unified-translator -provider cerebras -model llama3.1-70b: “The morning sun rose over the quiet village.” → “El sol de la mañana se alzó sobre el tranquilo pueblo.” — distinct from SambaNova’s output, proving a genuine per-provider model; Wave-3a, asciinema→agg→ffmpeg, ffmpeg 4.03s/4 frames + 11 extracted frames, ≥6 frames content-verified, start-frame ≠ output-frame per §11.4.107) | Confirmed:/Volumes/T7/Downloads/Recordings/helix_translate-provider-cerebras-translate-wave3a-20260616.mp4 | | Cloudflare | Cloudflare Workers AI client | Library | Implemented | Wired | /client/v4 Workers AI | Not-inventoried | Source-confirmed | PENDING | | SiliconFlow | SiliconFlow client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | Hyperbolic | Hyperbolic client | Library | Implemented | Wired | OpenAI-compatible wrapper (https://api.hyperbolic.xyz/v1, model meta-llama/Meta-Llama-3.1-70B-Instruct) | Not-inventoried | **PASS — real EN→ES translation captured** (via unified-translator -provider openai -base-url https://api.hyperbolic.xyz/v1 -model meta-llama/Meta-Llama-3.1-70B-Instruct: “The morning sun rose over the quiet village. An old fisherman walked along the shore...” → “Al amanecer, el sol salía sobre el tranquilo pueblo. Un anciano pescador caminaba por la

orilla...” — distinct verbs (salía/caminaba) from Fireworks’s output (ascendió/paseaba), proving a genuine per-provider model; Wave-3c, asciinema→agg→ffmpeg, ffprobe 8.01s/790×560 yuv420p + 16 extracted frames at fps=2, ≥6 frames content-verified, start-frame (input only) ≠ output-frame (Spanish) per §11.4.107) | Confirmed:/Volumes/T7/Downloads/Recordings/helix_translate-provider-hyperbolic-translate-wave3c-20260616.mp4 | | TogetherAI | TogetherAI client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | SambaNova | SambaNova client | Library | Implemented | Wired | OpenAI-compatible wrapper (https://api.sambanova.ai/v1, model Meta-Llama-3.1-70B-Instruct) | Not-inventoried | **PASS — real EN→ES translation captured** (via unified-translator -provider sambanova -model Meta-Llama-3.1-70B-Instruct: “The morning sun rose over the quiet village.” → “El sol matutino se alzó sobre el tranquilo pueblo.” — distinct from Cerebras’s output, proving a genuine per-provider model; Wave-3a, asciinema→agg→ffmpeg, ffprobe 4.39s/4 frames + 12 extracted frames, ≥6 frames content-verified, start-frame ≠ output-frame per §11.4.107) | Confirmed:/Volumes/T7/Downloads/Recordings/helix_translate-provider-sambanova-translate-wave3a-20260616.mp4 | | Kimi | Kimi (Moonshot) client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | Novita | Novita client | Library | Implemented | Wired | /v3/ openai-compatible wrapper | Not-inventoried | Source-confirmed | PENDING | | NLPCloud | NLPCloud client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | Upstage | Upstage (Solar) client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | Sarvam | Sarvam client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | Modal | Modal client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | PublicAI | PublicAI client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | NIA | NIA client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | Vulavula | Vulavula client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING |

pkg/translator — translation core

Component	Feature	Category	Implementation	Wiring	Real-use
Translator Core	Translator interface + BaseTranslator	Library	Implemented	Wired	Translate/TranslateWithProgress GetStats + base impl v caching & stats
Translator Core	Translation config	Library	Implemented	Wired	

Component	Feature	Category	Implementation	Wiring	Real-use
Translator Core	UniversalTranslator	Library	Implemented	Wired	TranslationConfig (provider/model/key/term/tokens/lang-pair/script) High-level ebook translator: lang detection, per-chapter translation, progress events

pkg/ebook — format parsers & writers

Component	Feature	Category	Implementation	Wiring	Real-use
ebook	Universal parser dispatcher	Library	Implemented	Wired	Multi-format parser (FB2/EPUB/DOCX/HTML/TXT/PDF)
ebook	FB2 parse	Library	Implemented	Wired	Parse FictionBook 2.0 → Book, rtf
ebook	FB2 write	Library	Implemented	Wired	Write Book → FB2 XML
ebook	EPUB parse	Library	Implemented	Wired	Parse EPUB (ZIP+OPF/NCX)
ebook	EPUB write	Library	Implemented	Wired	Write Book → EPUB 2.0 ZIP
ebook	DOCX parse	Library	Implemented	Wired	Parse .docx (archive/zip + encoding/xml)
ebook	HTML parse	Library	Implemented	Wired	Parse HTML → Book
ebook	TXT parse	Library	Implemented	Wired	Parse plain text (configurable 64k)
ebook	PDF text extraction	Library	Implemented	Wired	Extract text via ledongthuc/pdf
ebook	DOCX write (output)	Library	Implemented	Wired	Pure-Go OOXXML writer pkg/ebook/docx_wrt renders the in-memory Book to UOXML parts; XML specials auto-escape chardata. Wired into unified-translator (docx). Real DeepSeek EN→ES - Microsoft Word 2007+, well-formatted, the real translation. Guards: pkg/ebook/docx_wrt/subtest in cmd/unified-translator/ (proven)
ebook	PDF write (output)	Library	Implemented	Wired	PDF writer pkg/ebook/pdf_wrt renders the in-memory Book to UOXML parts; XML specials auto-escape chardata. Wired into unified-translator (pdf). Weasyprint to emit a VALID, CJK fonts can't render Cyrillic — weasyprint silent-glyph-drop bluff). Wired into generateOutput (case .pdf → e-book.pdf; main.go:1160); .pdf is a supported format

Component	Feature	Category	Implementation	Wiring	Real-use
					typed ErrWeasyPrintUnavailable (§11.4.3/§11.4.6). Guards: pkg/eb (TestPDFWriter_ProducesVal1 _UnavailableWeasyPrint)

pkg/fb2 — FB2-specific XML

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
fb2	FB2 parser (mixed-content)	Library	Implemented	Wired	Parse FB2 preserving paragraphs/ verses/ inline mixed content	Not-inventoried
fb2	FB2→Markdown converter	Library	Implemented	Wired	Convert FB2 → Markdown (author/ annotation/ epigraph/ poem/cite)	Not-inventoried

pkg/format

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
format	Format detector	Library	Implemented	Wired	Detect format by magic bytes/ extension/ content; ZIP disambiguation (EPUB/ DOCX/ AZW3)	Not-inventoried	Sou con

pkg/markdown — EPUB↔Markdown workflow

Component	Feature	Category	Implementation	Wiring	Real-use
markdown	EPUB→Markdown converter	Library	Implemented	Wired	EPUB→MD w/ YAML image extraction, GFM

Component	Feature	Category	Implementation	Wiring	Real-use
markdown	Markdown→EPUB converter	Library	Implemented	Wired	MD→EPUB 2.0 w/ OP tables/blockquotes
markdown	Simple workflow	Library	Implemented	Wired	ebook→markdown→tra pipeline w/ progress
markdown	Markdown translator	Library	Implemented	Wired	Line-by-line MD transla preserving formatting/c

pkg/script

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
script	Cyrillic↔Latin conversion	Library	Implemented	Wired	Bidirectional Serbian script conversion, context-aware digraphs	Not-inventoried

pkg/storage

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
storage	Storage interface	Library	Implemented	Wired	Abstract session + translation-cache persistence	Not-inventoried	So cor
storage	PostgreSQL backend	Library	Implemented	Wired	Pooled Postgres w/ UNIQUE-tuple dedup	Not-inventoried	So cor
storage	Redis backend	Library	Implemented	Wired	Redis cache, sha256-hashed tuple keys (collision-proof)	Not-inventoried	So cor
storage	SQLite backend	Library	Implemented	Wired	SQLite (optional SQLCipher) w/ UNIQUE-	Not-inventoried	So cor

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
storage	Cache-key hashing	Library	Implemented	Wired	tuple migrations Length-prefixed tuple encoding for collision-resistant keys	Not-inventoried	So cor

pkg/security

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Valida
security	JWT auth service	Library	Implemented	Wired	JWT generate/validate/refresh w/ timing-attack mitigation	Not-inventoried	Source confirm
security	API-key store	Library	Implemented	Wired	Thread-safe key validation w/ expiry & revocation	Not-inventoried	Source confirm
security	Rate limiter	Library	Implemented	Wired	Token-bucket per-key limiting w/ cleanup & stats	Not-inventoried	Source confirm
security	User auth service	Library	Implemented	Wired	Login/user-create w/ timing & enumeration mitigation	Not-inventoried	Source confirm

pkg/verification — quality verification & polishing

Component	Feature	Category	Implementation	Wiring	Real-use
verification	Heuristic verifier	Library	Implemented	Wired	Untranslated-block/HTML-artifact detection, quality sc
verification		Library	Implemented	Wired into unified-	Iterative multi-LLM polishing note-taking & consensus. N

Component	Feature	Category	Implementation	Wiring	Real-use
	Multi-pass polisher			translator (-multipass, d53e085)	reachable via the unified-translator opt-in -multipass flag (+ -multipass-passe -multipass-db; default O zero overhead/behavior cha when absent). NOTE: distin from -verify (the CLI's ov per-step output check) — -multipass invokes this p verification engine. Guac cmd/unified-translator/ multipass_wiring_test.go (httptest + real DeepSeek). §11.4.150 research: docs/ research/ 20260615_video_surfaced_ verification_multipass_wiri
verification	Book polisher	Library	Implemented	Wired	Multi-LLM per-dimension scoring & consensus valida
verification	Polishing reporter	Library	Implemented	Wired	Stats aggregation (consensu rates, issue/provider breakd
verification	Polishing database	Library	Implemented	Wired	SQLite persistence for mult sessions/pass records
verification	Literary notes	Library	Implemented	Wired	Character/tone/theme/ terminology/cultural note collection

internal/verifier — LLMsVerifier consumer integration

Consumer-side LLMsVerifier verification gate (client + scoring + discovery + selection). Source: internal/verifier/pipeline.go.

Component	Feature	Category	Implementation	Wiring	Real-use
internal/verifier	Affirmative-response hard-gate	Library	Implemented	Wired (verification pipeline)	Correctness fix (commit 97a8afd): a model with affirmative_response= is now a HARD disqualifi — rejected by the verification gate regardless of other scores

Component	Feature	Category	Implementation	Wiring	Real-use

pkg/preparation — pre-translation analysis

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	V
preparation	Preparation coordinator	Library	Implemented	Wired	Multi-pass content analysis (character/terminology/cultural)	Not-inventoried	S c
preparation	Preparation-aware translator	Library	Implemented	Wired	Wraps base translator w/ analysis phase + context passing	Not-inventoried	S c
preparation	Prompt builder	Library	Implemented	Wired	Context-aware analysis prompts refined across passes	Not-inventoried	S c
preparation	Domain types	Library	Implemented	Wired	Content-classification/character/cultural/terminology structs	Not-inventoried	S c
preparation	Utilities	Library	Implemented	Wired	JSON serialization + human-readable summaries	Not-inventoried	S c

pkg/coordination — multi-LLM coordination

Component	Feature	Category	Implementation	Wiring	Real-use	Te
coordination		Library	Implemented	Wired		No inv

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
coordination	Multi-LLM coordinator	Library	Implemented	Wired	Multiple LLM instances w/ round-robin, retry, distributed fallback	Not-inventoried
	Multi-LLM translator wrapper				Adapts coordinator to translator.Translator interface	

pkg/distributed — SSH distributed workers

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
distributed	Coordinator (distribute + aggregate)	Library	Implemented	Wired	Distributes work across remote LLM instances, round-robin, failover to local	Not-inventoried
distributed	SSH connection pool	Library	Implemented	Wired	Pooled SSH connections w/ idle-timeout cleanup	Not-inventoried
distributed	Worker pairing/discovery	Library	Implemented	Wired	Discover/pair/health-check remote translator services	Not-inventoried
distributed	Fallback / graceful degradation	Library	Implemented	Wired	Retry w/ backoff, degraded-mode entry/exit, recovery tracking	Not-inventoried
distributed	Version manager	Library	Implemented	Wired	Version check/update/rollback, signed packages, backups,	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
distributed	Performance instrumentation	Library	Implemented	Wired	drift alerts (no LLM) Conn pooling, TTL result cache, circuit breaker, batch processing SSH host-key verify, TLS validation, network ACLs, security event log	Not-inventoried
distributed	Security hardening	Library	Implemented	Wired	Init/discover/pair/version-manage + translation delegation	Not-inventoried
distributed	Manager / orchestration	Library	Implemented	Wired		Not-inventoried

pkg/sshworker

Obsolete-Details: Since=bridge phase-2 R-2/R-4 (2026-06-16); Reason=feature-removed (local-runtime + SSH-local translation removed; default path now sources the LLMsVerifier bridge — no local llama.cpp/Ollama, no SSH-local worker); Superseding-item=pkg/bridge (LLMsVerifier→agent bridge); Operator-approval=2026-06-15 D1/D2 (forbid all local runtimes) + R-1 redirect + the R-4 “keep distributed API, remove SSH-local” decision (2026-06-16); Triple-check evidence=source files removed (FACT: `git ls-files` empty for the removed `cmd/*` dirs + `pkg/sshworker` + `pkg/modelsbridge` + `ollama.go/llamacpp*.go`) + CM-NO-LOCAL-RUNTIME gate PASS. The KEPT distributed-API path is NOT Obsolete. | Component | Feature | Category | Implementation | Wiring | Real-use | Tests | Validation | Video-confirmation | |-----|-----|-----|-----|-----|-----|-----|-----|-----|-----| sshworker | SSH worker | Library | **Obsolete (→ Fixed.md)** | Wired | SSH exec, file up/download, codebase sync, key generation | Not-inventoried | Source-confirmed | N/A | | sshworker | Monitored worker | Library | **Obsolete (→ Fixed.md)** | Wired | Event-monitored progress tracking for long commands | Not-inventoried | Source-confirmed | N/A |

pkg/deployment

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
deployment	Deployment orchestrator	Library	Implemented	Wired	Automated multi-host instance deployment	Not-inver
deployment	Docker orchestrator	Library	Implemented	Wired	Generate/manage docker-compose for multi-service deploy	Not-inver
deployment	SSH deployer	Library	Implemented	Wired	SSH-based deploy w/ validation & error handling	Not-inver
deployment	Network discovery	Library	Implemented	Wired	UDP service discovery/ broadcast	Not-inver
deployment	API comm logger	Library	Implemented	Wired	Retrofit-style REST request/response logging between nodes	Not-inver
deployment	Deployment types	Library	Implemented	Wired	DeploymentPlan/ Status/NetworkService/ APICommunicationLog	Not-inver

pkg/grpc

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
grpc	gRPC TranslationService	Library	Implemented	Wired	gRPC server: sessions, event streaming, provider registry, concurrency	Not-inventorie
grpc	Core translator pipeline	Library	Implemented	Wired	Translation job exec w/ step tracking + progress via llm factory	Not-inventorie

pkg/api (library layer)

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Va
api	REST API server	Library	Implemented	Wired	HTTP server: auth, CSRF,	Not-inventoried	Sc co

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
api	Translation endpoints	Library	Implemented	Wired	rate-limit, routing text/FB2/ ebook translate, batch ops, lang/provider listing	Not-inventoried	Sc co
api	Batch translation handlers	Library	Implemented	Wired	Batch requests w/ queue, progress, concurrency	Not-inventoried	Sc co
api	Verifier endpoints	Library	Implemented	Wired	LLMsVerifier model verification/ scoring/ selection; / verified-models	Not-inventoried	Sc co

pkg/events, websocket, batch, progress, report, logger, hardware, hash

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
events	Event pub/sub bus	Library	Implemented	Wired	Thread-safe pub/sub w/ subscription IDs + type filtering	Not-inventoried	Sou con
websocket	WebSocket hub	Library	Implemented	Wired	Real-time session event broadcast to dashboards	Not-inventoried	Sou con
batch	Batch processor	Library	Implemented	Wired	File/dir translation w/ concurrency & progress	Not-inventoried	Sou con
progress	Progress tracker	Library	Implemented	Wired	Chapter/section progress w/ ETA	Not-inventoried	Sou con
report		Library	Implemented	Wired			

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
	Report generator				Session reporting (issues/warnings/logs/results)	Not-inventoried	Sou
logger	Structured logger	Library	Implemented	Wired	Text/JSON logging w/ levels	Not-inventoried	Sou
hardware	Hardware detector	Library	Implemented	Wired	CPU/RAM/GPU detection + model-size estimation	Not-inventoried	Sou
hash	Codebase hash generator	Library	Implemented	Wired	SHA256 of source/config/deps/binaries	Not-inventoried	Sou

pkg/language

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Validation
language	Language registry / heuristic detector	Library	Implemented	Wired	ISO 639-1 registry + heuristic detection (18+ langs)	Not-inventoried	Source-confirmed
language	LLM language detector	Library	Implemented	Wired	LLM-driven language detection	Not-inventoried	Source-confirmed

pkg/models & modelsbridge

Obsolete-Details: Since=bridge phase-2 R-2/R-4 (2026-06-16); Reason=feature-removed (local-runtime + SSH-local translation removed; default path now sources the LLMsVerifier bridge — no local llama.cpp/Ollama, no SSH-local worker); Superseding-item=pkg/bridge (LLMsVerifier→agent bridge); Operator-approval=2026-06-15 D1/D2 (forbid all local runtimes) + R-1 redirect + the R-4 “keep distributed API, remove SSH-local” decision (2026-06-16); Triple-check evidence=source files removed (FACT: `git ls-files` empty for the removed cmd/* dirs +

pkg/sshworker + pkg/modelsbridge + ollama.go/llamacpp*.go) + CM-NO-LOCAL-RUNTIME gate PASS. The KEPT distributed-API path is NOT Obsolete. | Component | Feature | Category | Implementation | Wiring | Real-use | Tests | Validation | Video-confirmation | |---| | models | Model registry | Library | Implemented | Wired | Translation-model metadata registry (50+ models) | Not-inventoried | Source-confirmed | N/A | | models | Model downloader | Library | Implemented | Wired | HTTP model download w/ caching + sha256 verification | Not-inventoried | Source-confirmed | N/A | | models | User domain | Library | Implemented | Wired | User struct, bcrypt hashing, in-memory repo | Not-inventoried | Source-confirmed | N/A | | models | Error definitions | Library | Implemented | Wired | Domain error constants | Not-inventoried | Source-confirmed | N/A | | models | Verifier registry | Library | Implemented | Wired | LLMsVerifier-backed registry w/ thresholds & filtering | Not-inventoried | Source-confirmed | PENDING | | modelsbridge | Models bridge | Library | **Obsolete (→ Fixed.md)** | Wired | Adapter mapping models.LLMRequest/Response → llm.LLMClient | Not-inventoried | Source-confirmed | N/A |

pkg/bridge — LLMsVerifier→component+agent bridge

In-process bridge (no local models) that selects the best LLMsVerifier-scored model and routes component/agent calls to it; backs cmd/model-bridge CLI + MCP. Source: pkg/bridge/bridge.go, commit ab1bed3 (+ routing mutation seam a5860b2).

Component	Feature	Category	Implementation	Wiring
bridge	Best-model selection	Library	Implemented	Wired (via cmd/model-bridge CLI+MCP)
bridge	Invoke routes to BestModel	Library	Implemented	Wired
bridge	BestTranslator (single strongest translator + fallback chain)	Library	Implemented	Wired (R-1 redirect seam)

Component	Feature	Category	Implementation	Wiring
bridge	BestTranslatorFunc (context-only single-translator closure)	Library	Implemented	Wired (R-1 redirect seam)
bridge	EnsembleFactory (provider-diverse ensemble seam)	Library	Implemented	Wired (R-1 ensemble redirect)
bridge	ProviderDiverseTranslators	Library	Implemented	Wired (backs EnsembleFactory)
bridge	ProviderDiverseClients	Library	Implemented	Wired
bridge	ProviderDiverseModels	Library	Implemented	Wired (backs the ProviderDiverse* set)

Component	Feature	Category	Implementation	Wiring
bridge	BestClient (single strongest low-level client)	Library	Implemented	Wired

pkg/version

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Validation
version	App version	Library	Implemented	Wired	Singleton version constant (2.3.0)	Not-inventoried	Source confirmed
version	Codebase hasher	Library	Implemented	Wired	SHA256 codebase versioning over dirs/extensions w/ exclude patterns	Not-inventoried	Source confirmed

pkg/challenge_runner — anti-bluff test harness (uses digital.vasic.challenges)

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
challenge_runner	Challenge orchestrator	Library	Implemented	Wired	Coordinates 18+ integration challenges, parallel exec + filtering	Not-inventoried
challenge_runner	Build & format-detection challenges	Library	Implemented	Wired	Build-verification + FB2/EPUB/TXT/HTML/PDF/DOCX format-detection challenges	Not-inventoried
challenge_runner	Infrastructure challenges	Library	Implemented	Wired	Event-bus + other infra challenges	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
challenge_runner	Provider challenges	Library	Implemented	Wired	(run go test) Verify LLM provider factory/ interface compile+load (build, not live calls)	Not-inventori

3. Web Dashboards & Endpoints

Web Dashboards

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
monitor.html	WebSocket connection status	Web	Implemented	Wired	Green/red pulse dot + Connected/Disconnected indicator	Not-inver
monitor.html	Session/Client ID display	Web	Implemented	Wired	Static fields (demo-session-123, web-demo)	Not-inver
monitor.html	Overall progress bar + percent	Web	Implemented	Wired	Driven by WS translation_progress/step_completed events	Not-inver
monitor.html	Current step / step message panel	Web	Implemented	Wired	Shows current step + message from WS	Not-inver
monitor.html	Duration timer	Web	Implemented	Wired	Client-side elapsed clock	Not-inver
monitor.html	Status badge	Web	Implemented	Wired	Pending/Running/Completed/Failed	Not-inver
monitor.html	Event log	Web	Implemented	Wired	Scrolling list of received WS events	Not-inver
monitor.html	Control Panel: Simulate Progress	Web	Implemented	Wired	Locally fakes & echoes progress events over WS (demo only)	Not-inver

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
monitor.html	Clear Log / Disconnect / Reconnect WS	Web	Implemented	Wired	Local log clear + WS connect/disconnect to ws://localhost:8090/ws	Not-inver
enhanced-monitor.html	Stats header (Active/Completed/Errors)	Web	Implemented	Wired	Aggregated counters from WS-tracked sessions	Not-inver
enhanced-monitor.html	Control panel: Connect/Disconnect/Clear	Web	Implemented	Wired	WS to ws://localhost:8090/ws?client_id=dashboard	Not-inver
enhanced-monitor.html	Monitor specific session	Web	Implemented	Wired	Input + button to filter UI to a given session_id	Not-inver
enhanced-monitor.html	Worker Information panel	Web	Implemented	Wired	Host:port, model, capacity from WS worker_info	Not-inver
enhanced-monitor.html	Current Session Progress panel	Web	Implemented	Wired	Progress bar, step, current/total items, duration, status	Not-inver
enhanced-monitor.html	Live progress chart (Chart.js)	Web	Implemented	Wired	Progress % over time, last 20 points	Not-inver
enhanced-monitor.html	Event log	Web	Implemented	Wired	Color-coded WS event stream	Not-inver
enhanced-monitor.html	Session History table	Web	Implemented	Wired	Last 10 sessions: id, worker, status, duration, progress, start	Not-inver
dashboard.html	Stats cards (Total/Completed/Running/Failed)	Web	Implemented	Wired	Polls GET /api/v1/translations every 5s	Not-inver
dashboard.html	Active Sessions table	Web	Implemented	Wired	Lists sessions w/ provider, status, progress, duration	Not-inver
dashboard.html	Session Details view	Web	Implemented	Wired	Progress bar, steps, generated files via GET /api/v1/translations/:id	Not-inver
dashboard.html		Web	Implemented	Wired	Connects ws://localhost:8080/ws?session_id=...;	Not-inver

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
	Live progress via WebSocket				progress_update/step_*	
dashboard.html	New Translation modal	Web	Implemented	Wired	File input, provider selector, source/target lang, API-key/SSH/llama fields (UI only)	Not-inver
dashboard.html	Start Translation (submit)	Web	Implemented	Wired	POST /api/v1/translations with provider config — triggers backend translation	Not-inver
dashboard.html	Cancel session	Web	Implemented	Wired	DELETE /api/v1/translations/:id	Not-inver

Web Dashboard backend handler — pkg/api/dashboard.go

Backend that serves the dashboard page (/, /dashboard, /monitor) and the translations endpoints. Was previously UNROUTED (returned 404); now wired so the dashboard genuinely translates for end users. Source: pkg/api/dashboard.go, commit f6ba5cc.

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
api/ dashboard.go	Dashboard page + translations endpoints	Web	Implemented	Wired (was Unwired/404 pre-fix)	Serves the dashboard HTML at /, /dashboard, /monitor and the GET/POST /api/v1/translations path; Start-Translation now runs a real translation through the wired UI path (no longer a 404)	Covered + 5 RED→GR guard dashboard (default GR asserts page real translat RED_MODE= reproduces fix 404) — go test -pkg/api G this session

REST API — pkg/api/server.go (standalone Server, gin)

Component	Feature	Category	Implementation	Wiring	Real-use
api/ server.go	GET / health	API	Implemented	Wired	Status + translator name
api/ server.go	POST / api/ translate	API	Implemented	Wired	translator.Translate(...)
api/ server.go	GET /api/ languages	API	Implemented	Wired	Static language list
api/ server.go	GET /api/ stats	API	Implemented	Wired	Returns hardcoded zeros (not live metrics)
api/ server.go	POST / api/ upload	API	Implemented	Wired	Save uploaded file, detect format
api/ server.go	POST / api/batch	API	Partial	Wired	Returns batch_id queued only; performs NO translation

REST API — pkg/api/handler.go (primary Handler, gin)

Component	Feature	Category	Implementation	Wiring	Real-use
api/ handler.go	GET /health, /, /api/v1/version, /api/v1/providers, /api/v1/stats	API	Implemented	Wired	Static/info cache stats
api/ handler.go	GET /ws	API	Implemented	Wired	WebSockets to wsHub
api/ handler.go	POST /api/v1/translate	API	Implemented	Wired	Distributed Translate target_lang was hardcoded

Component	Feature	Category	Implementation	Wiring	Real-use
api/ handler.go	POST /api/v1/translate/fb2	API	Implemented	Wired	Parse ebook
api/ handler.go	POST /api/v1/translate/batch	API	Implemented	Wired	Loops text
api/ handler.go	POST /api/v1/convert/script	API	Implemented	Wired	Cyrillic ^[2]
api/ handler.go	GET /api/v1/status/:session_id	API	Implemented	Wired	Returns ha a live sess
api/ handler.go	GET /api/v1/languages	API	Implemented	Wired	Static list language.C
api/ handler.go	POST /api/v1/translate/validate	API	Implemented	Wired	Validates n length); no
api/ handler.go	POST /api/v1/preparation/ analyze	API	Stub	Wired	File stat/co analysis, m
api/ handler.go	GET /api/v1/preparation/ result/:session_id	API	Stub	Wired	Returns m
api/ handler.go	POST /api/v1/translate/ebook	API	Stub	Wired	Emits star does NOT
api/ handler.go	POST /api/v1/translate/ cancel/:session_id	API	Implemented	Wired	Emits can cancelle
api/ handler.go	GET /api/v1/distributed/status	API	Implemented	Wired	dm.GetSt
api/ handler.go	POST /api/v1/distributed/ workers/discover	API	Implemented	Wired	SSH work
api/ handler.go	POST /api/v1/distributed/ workers/:worker_id/pair	API	Implemented	Wired	Pair work
api/ handler.go	DELETE /api/v1/distributed/ workers/:worker_id/pair	API	Implemented	Wired	Unpair wo
api/ handler.go	POST /api/v1/distributed/ translate	API	Implemented	Wired	dm.Trans
api/ handler.go	POST /api/v1/update/ {upload,apply,rollback}	API	Implemented	Wired	Worker se install/rol
api/ handler.go	GET /api/v1/monitoring/version/ {metrics,alerts,health,dashboard}	API	Implemented	Wired	Version-sy
api/ handler.go	POST /api/v1/monitoring/ version/drift-check	API	Implemented	Wired	Trigger dr

Component	Feature	Category	Implementation	Wiring	Real-use
api/ handler.go	GET /api/v1/monitoring/version/ alerts/history	API	Implemented	Wired	Alert histo
api/ handler.go	POST /api/v1/monitoring/ version/alerts/:alert_id/ acknowledge	API	Implemented	Wired	Ack alert
api/ handler.go	POST /api/v1/monitoring/ version/alerts/channels/ {email,webhook,slack}	API	Implemented	Wired	Add alert c
api/ handler.go	GET /api/v1/monitoring/version/ dashboard.html	Web	Implemented	Wired	Serve emb
api/ handler.go	POST /api/v1/auth/login, /auth/ token	API	Implemented	Wired	JWT auth
api/ handler.go	GET /api/v1/profile	API	Implemented	Wired	Return cla middleware

REST API — pkg/api/batch_handlers.go (under /api/v1)

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
api/ batch_handlers.go	POST / api/v1/ translate/ string	API	Implemented	Wired	NewLLMTranslator + Translate	Not- inven
api/ batch_handlers.go	POST / api/v1/ translate/ directory	API	Implemented	Wired	Batch processor translates each file	Not- inven

REST API — pkg/api/verifier_handlers.go (LLMsVerifier, under /api/v1)

Component	Feature	Category	Implementation	Wiring	Real-use
api/ verifier_handlers.go	GET /api/ v1/verified- models	API	Implemented	Wired (only when LLMsVerifier enabled)	List verified models (LLMsVerifier API call). GAP (§11.4.6): LLMsVerifier is DISABLED in the current config, so the route is not mounted and returns HTTP 404. Not a defect — feature- disabled-by- config; needs a verifier- enabled config to confirm
api/ verifier_handlers.go	GET /api/ v1/verified- models/:id	API	Implemented	Wired	Get one verified model
api/ verifier_handlers.go	GET /api/ v1/ verification- status	API	Implemented	Wired	Ping LLMsVerifier
api/ verifier_handlers.go	POST /api/ v1/ verification/ refresh	API	Implemented	Wired	Refresh scores
api/ verifier_handlers.go	GET /api/ v1/ providers/ verified	API	Implemented	Wired	Aggregated verified providers
api/ verifier_handlers.go	POST /api/ v1/ translate- with- verification	API	Stub	Wired	Selects/ verifies a model, returns metadata + text preview; does NOT translate

REST API — cmd/api-server/main.go (gRPC-proxying REST, :8080)

Component	Feature	Category	Implementation	Wiring	Real-use
api-server	POST /api/v1/translations	API	Implemented	Wired	Proxies grpcClient.StartTranslating
api-server	GET /api/v1/translations/:session_id	API	Implemented	Wired	Get status (proxy)
api-server	GET /api/v1/translations	API	Implemented	Wired	List (proxy)
api-server	DELETE /api/v1/translations/:session_id	API	Implemented	Wired	Cancel (proxy)
api-server	GET /api/v1/translations/:session_id/stream	API	Implemented	Wired	WS proxy of gRPC StreamTranslationP
api-server	GET /api/v1/providers	API	Implemented	Wired	Get providers (prox
api-server	GET /api/v1/health	API	Implemented	Wired	Derived from gRPC state
api-server	GET /api/v1/metrics	API	Implemented	Wired	Returns static/zero m (not live)
api-server	GET /ws	Web	Implemented	Wired	WS upgrade, register wsHub (requires session_id)
api-server	GET /static/*	Web	Implemented	Wired	Static file server (./static)
api-server	GET /	Web	Implemented	Wired	Render dashboard.h template

WebSocket — hub & fan-out

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Valid
websocket/hub.go	/ws endpoint (package mux)	Infra	Implemented	Wired	WS upgrade; reads client_id + session_id	Not-inventoried	Source confi
websocket/hub.go	Event fan-out	Infra	Implemented	Wired	Subscribes to ALL bus events; marshals events.Event JSON; per-	Not-inventoried	Source confi

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Valid
					session filter; Broadcast()		

gRPC — pkg/grpc/server.go (service TranslationService)

Component	Feature	Category	Implementation	Wiring	Real-use
grpc	StartTranslation	gRPC	Implemented	Wired	Reserves session spawns runTrans → translator.Transla
grpc	GetTranslationStatus	gRPC	Implemented	Wired	Snapshot session translator.GetSta
grpc	ListTranslations	gRPC	Implemented	Wired	Iterate sessions
grpc	CancelTranslation	gRPC	Implemented	Wired	Cancel ctx + translator.Cancel
grpc	StreamTranslationProgress	gRPC	Implemented	Wired	Stream session progress events
grpc	GetProviders	gRPC	Implemented	Wired	Static ProviderRegistry (openai, anthropi ssh)
grpc	SubscribeEvents	gRPC	Implemented	Wired	Stream filtered SystemEvents fro bus

CORS / Middleware / Auth (Infra)

Component	Feature	Category	Implementation	Wiring	Real-use
cmd/server	corsMiddleware	Infra	Implemented	Wired	Driven by cfg.Security.CORSOrig reflects allowlisted Or + Allow-Credentials; wildcard * no-credenti OPTIONS→204
cmd/server	rateLimitMiddleware	Infra	Implemented	Wired	Per-ClientIP via security.RateLimiter; 4 on exceed
	authMiddleware	Infra	Implemented	Wired	Bearer JWT validation applied to protected

Component	Feature	Category	Implementation	Wiring	Real-use
pkg/api/handler.go					subgroup, gated by Security.EnableAuth
pkg/api/server.go	authMiddleware (standalone)	Infra	Implemented	Wired	X-API-Key header == config.Security.APIKey Security.RequireAuth
pkg/grpc/server.go	message-size limits	Infra	Implemented	Wired	4 MB recv/send msg-s limits; no interceptors/auth/rate-limit register

4. Owned Submodules

challenges/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
challenges	Challenge core engine	Submodule	Implemented	Wired	Challenge interface, Config, Result, BaseChallenge, ProgressReporter (liveness)	Not-inver
challenges	Registry + dependency ordering	Submodule	Implemented	Wired	Challenge registration, Kahn topological sort by deps	Not-inver
challenges	Runner / execution engine	Submodule	Implemented	Wired	Sequential/parallel/ pipeline execution, liveness/stall monitoring	Not-inver
challenges	Assertion engine	Submodule	Implemented	Wired	16 built-in evaluators (not_empty, contains, min_count, max_latency...) + custom	Not-inver
challenges	Report generation	Submodule	Implemented	Wired	Markdown / JSON / HTML reporters	Not-inver
challenges	Live monitoring	Submodule	Implemented	Wired	WebSocket dashboard, progress liveness	Not-inver
challenges	Metrics	Submodule	Implemented	Wired	Prometheus-compatible challenge metrics	Not-inver
challenges	HTTP client	Submodule	Implemented	Wired	Generic REST client with JWT auth, functional options	Not-inver

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
challenges	Challenge bank loader	Submodule	Implemented	Wired	Load challenge definitions from JSON/YAML	Not-inventoried
challenges	Plugin system	Submodule	Implemented	Wired	Custom challenge types + assertion plugins	Not-inventoried
challenges	Infra bridge	Submodule	Implemented	Wired	Bridge to digital.vasic.containers for test infra	Not-inventoried
challenges	User-flow automation	Submodule	Implemented	Wired	8 adapter interfaces / 21 impls (Playwright, ADB, Tauri, HTTP, gRPC, WS, Gradle, Cargo, npm), 19 templates, 12 evaluators	Not-inventoried
challenges	userflow-runner CLI	Submodule	Implemented	Wired	CLI runner for user-flow challenges	Not-inventoried
challenges	panoptic / i18n / env / logging	Submodule	Implemented	Wired	Observation, i18n, env-var redaction, structured logging	Not-inventoried
challenges	anti_bluff lib	Submodule	Implemented	Wired	Shell anti-bluff helper library	Not-inventoried

containers/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
containers	boot manager	Submodule	Implemented	Wired	On-demand infra boot/start orchestration	Not-inventoried
containers	compose	Submodule	Implemented	Wired	docker/podman compose file orchestration	Not-inventoried
containers	health checking	Submodule	Implemented	Wired	TCP/HTTP health-check primitives	Not-inventoried
containers	runtime auto-detect	Submodule	Implemented	Wired	Docker/ Podman/ Kubernetes runtime detection	Not-inventoried
containers	endpoint	Submodule	Implemented	Wired	ServiceEndpoint builder	Not-inventoried
containers	lifecycle	Submodule	Implemented	Wired		

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					Container lifecycle management	Not-inventoried
containers	service discovery / registry	Submodule	Implemented	Wired	Service discovery + registry	Not-inventoried
containers	orchestrator / scheduler	Submodule	Implemented	Wired	Orchestration + scheduling	Not-inventoried
containers	emulator suite	Submodule	Implemented	Wired	Android/device emulator boot/matrix/cleanup/canary	Not-inventoried
containers	genymotion	Submodule	Implemented	Wired	Genymotion device support	Not-inventoried
containers	vm-matrix	Submodule	Implemented	Wired	VM matrix orchestration	Not-inventoried
containers	distributed build/test	Submodule	Implemented	Wired	Distributed build + test commands	Not-inventoried
containers	crossbuild	Submodule	Implemented	Wired	Cross-platform build	Not-inventoried
containers	ctop	Submodule	Implemented	Wired	Container top / monitoring TUI	Not-inventoried
containers	network / volume	Submodule	Implemented	Wired	Network + volume primitives	Not-inventoried
containers	supporting primitives	Submodule	Implemented	Wired	cache/metrics/monitor/event/policy/remote/lazyservice/brokertest/envconfig	Not-inventoried

doc_processor/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
doc_processor	Document loader	Submodule	Implemented	Wired	Loads Markdown / YAML / HTML docs (10 MB cap)	Not-inventoried
doc_processor	Feature-map builder	Submodule	Implemented	Wired	Builds structured FeatureMap; heuristic extraction + LLM Enrich	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
doc_processor	LLMAgent interface	Submodule	Implemented	Wired	Summarize/ ExtractFeatures/ ClassifyFeature/ InferScreens/ GenerateTestSteps	Not-inven
doc_processor	Coverage tracker	Submodule	Implemented	Wired	Verification coverage tracking by platform/ category	Not-inven
doc_processor	DocGraph	Submodule	Implemented	Wired	Doc node/edge graph	Not-inven
doc_processor	Config / i18n	Submodule	Implemented	Wired	Configuration + i18n	Not-inven
doc_processor	docprocessor CLI	Submodule	Implemented	Wired	CLI running feature extraction against a docs dir	Not-inven
doc_processor	archdoc (internal)	Submodule	Implemented	Wired	Architecture-doc helper (LLM-use UNCONFIRMED)	Not-inven

docs_chain/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
docs_chain	sync command	Submodule	Implemented	Wired	Propagate changes through chain (regenerate + re-export atomically)	Not-inventori
docs_chain	verify command	Submodule	Implemented	Wired	Deterministic CI/ pre-build gate (chain in sync)	Not-inventori
docs_chain	doctor command	Submodule	Implemented	Wired	Diagnostics	Not-inventori
docs_chain	graph command	Submodule	Implemented	Wired	Show DAG / topo order	Not-inventori
docs_chain	watch daemon	Submodule	Implemented	Wired	fsnotify-driven continuous sync	Not-inventori
docs_chain	exec: transforms	Submodule	Implemented	Wired	Shell-out transforms (pandoc md→html/docx, weasyprint html→pdf)	Not-inventori

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
docs_chain	content-hash engine	Submodule	Implemented	Wired	Content-hash incremental recompute; roster/corpus fingerprint (§11.4.86)	Not-inventori
docs_chain	DAG + recompute	Submodule	Implemented	Wired	Node/Edge/Graph, cycle detection, Kahn topo, early-cutoff, sync-edge conflict	Not-inventori
docs_chain	adapters	Submodule	Implemented	Wired	FileAdapter, SQLiteAdapter (pure-Go modernc), DERIVED pandoc/weasyprint (ToolAbsentError)	Not-inventori
docs_chain	orchestrator	Submodule	Implemented	Wired	Atomic run: staged writes, SQLite-txn commit + rollback	Not-inventori
docs_chain	DB builtins	Submodule	Implemented	Wired	Bidirectional md-to-sqlite / sqlite-to-md (row-level drift); colorize-html	Not-inventori
docs_chain	config + state	Submodule	Implemented	Wired	YAML multi-context loader; state.json	Not-inventori

llm_orchestrator/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
llm_orchestrator	Agent interface + AgentPool	Submodule	Implemented	Wired	Thread-safe pool, capability matching, acquire/release, health	Not-invent
llm_orchestrator	CLI agent adapters	Submodule	Implemented	Wired	OpenCode, ClaudeCode, Gemini, Junie, QwenCode	Not-invent

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					adapters (BaseAdapter)	
llm_orchestrator	Transport / protocol	Submodule	Implemented	Wired	PipeTransport (JSON-line stdin/stdout) + FileTransport, protocol types	Not-invent
llm_orchestrator	Response parser	Submodule	Implemented	Wired	Parses agent JSON-line responses	Not-invent
llm_orchestrator	HealthMonitor + CircuitBreaker	Submodule	Implemented	Wired	Health tracking + fault tolerance over agents	Not-invent
llm_orchestrator	Config / i18n	Submodule	Implemented	Wired	Pool/agent config; i18n	Not-invent
llm_orchestrator	orchestrator CLI	Submodule	Implemented	Wired	CLI entry for orchestration	Not-invent

llm_provider/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
llm_provider	LLMProvider interface	Submodule	Implemented	Wired	Complete/ CompleteStream/ HealthCheck/ GetCapabilities/ ValidateConfig	Not-inver
llm_provider	Circuit breaker	Submodule	Implemented	Wired	closed/open/half-open fault tolerance	Not-inver
llm_provider	Health monitor	Submodule	Implemented	Wired	healthy/ degraded/ unhealthy/ unknown	Not-inver
llm_provider	Retry logic	Submodule	Implemented	Wired	Exponential backoff + jitter, HTTP-status detection	Not-inver
llm_provider	Model discovery	Submodule	Implemented	Wired	Provider/model discovery (HTTP/endpoint-based)	Not-inver
llm_provider	API-key management	Submodule	Implemented	Wired	API key handling	Not-inver
llm_provider		Submodule	Implemented	Wired	Shared HTTP client, request/	

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
	HTTP client + models + i18n				response models, i18n	Not- inver
					ai21, anthropic, cerebras, chutes, claude, cloudflare, codestral, cohere, deepseek, fireworks, gemini, generic, githubmodels, groq, huggingface, hyperbolic, junie, kilo, kimi, mistral, modal, nia, nlpcloud, novita, nvidia, ollama, openai, openrouter, perplexity, publicai, qwen, replicate, sambanova, sarvam, siliconflow, together, upstage, venice, vulavula, xai, zai, zen, zhipu (source llm_provider/ pkg/ providers/ <name>/)	
llm_provider	Provider implementations (43)	Submodule	Implemented	Wired		Not- inver

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
-----------	---------	----------	----------------	--------	----------	-------

llms_verifier/

Component	Feature	Category	Implementation	Wiring	Real-use	Te
llms_verifier	Model verification	Submodule	Implemented	Wired	“Do you see my code?” + existence/responsiveness/latency/streaming/function-calling/vision/embeddings tests	No inv
llms_verifier	Provider adapters	Submodule	Implemented	Wired	OpenAI, Anthropic, Cohere, Groq, Together, Mistral, xAI, Replicate, DeepSeek, Cerebras, Cloudflare, SiliconFlow (12+)	No inv
llms_verifier	Scoring engine	Submodule	Implemented	Wired		

Component	Feature	Category	Implementation	Wiring	Real-use	Te
					Quality scoring system (LLM-use UNCONFIRMED)	No inv
llms_verifier	Capabilities detection	Submodule	Implemented	Wired	Per-model capability discovery	No inv
llms_verifier	Failover / health monitoring	Submodule	Implemented	Wired	Real-time monitoring + intelligent failover (probes call LLM)	No inv
llms_verifier	AI analytics / recommendations	Submodule	Implemented	Wired	AI-powered insights, trend analysis, model recommendations	No inv
llms_verifier	RAG / context mgmt	Submodule	Implemented	Wired	Vector DB semantic search, LLM-powered summarization	No inv
llms_verifier	LLMOps / self-improve / benchmark	Submodule	Implemented	Wired	LLM operations, self-improvement loop, benchmarking	No inv
llms_verifier	Messaging	Submodule	Implemented	Wired	Notification/messaging integration (Splunk/DataDog/ etc.)	No inv
llms_verifier	suffix / verified export	Submodule	Implemented	Wired	Mandatory branding suffix + verified-only config export	No inv
llms_verifier	Auth / security / API / API-keys	Submodule	Implemented	Wired	LDAP/SSO/ SAML/OIDC, SQLCipher, REST API, key mgmt	No inv
llms_verifier	Multi-platform clients	Submodule	Implemented	Wired	CLI, TUI, Web, Desktop, Mobile (UI; LLM-backed)	No inv
llms_verifier	helixqa / cliagents / crush / opencode	Submodule	Implemented	Wired	Integration packages (LLM-use UNCONFIRMED)	No inv
llms_verifier	scheduler / partners / bigdata / multimodal / performance	Submodule	Implemented	Wired	Scheduling, partners, big-data, multimodal, perf (multimodal likely LLM)	No inv
llms_verifier	main CLI + ACP	Submodule	Implemented	Wired		

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					Main CLI; ACP (Agent Client Protocol)	No inv

security/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
security	Content guardrails	Submodule	Implemented	Wired	Rule pipeline (MaxLength, ForbiddenPatterns, RequireFormat) w/ severities	Not-invento
security	PII detection + redaction	Submodule	Implemented	Wired	Email/Phone/SSN/CreditCard(Luhn)/IPv4 detection; Mask/Hash/Remove	Not-invento
security	Content filtering	Submodule	Implemented	Wired	Composable chain: Length/Pattern/Keyword filters	Not-invento
security	Policy enforcement	Submodule	Implemented	Wired	Named policies, 8 condition operators, Allow/Deny/Audit + audit trail	Not-invento
security	Vulnerability scanner	Submodule	Implemented	Wired	Findings/reports, severity filtering + merging	Not-invento
security	HTTP security headers	Submodule	Implemented	Wired	Middleware: CSP/HSTS/X-Frame-Options + 7 standard headers	Not-invento
security	Secure storage	Submodule	Implemented	Wired	AES-256-GCM encrypted file KV store, Argon2id KDF	Not-invento
security	Privilege-escalation scanner	Submodule	Implemented	Wired	Probes /proc/self/status, capability bitmasks, namespace cgroups	Not-invento
security	SSRF denial	Submodule	Implemented	Wired	Denies SSRF on caller's outbound HTTP client	Not-invento

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
security	E2EE	Submodule	Implemented	Wired	End-to-end encryption	Not-invented
security	Attestation / GPU attestation	Submodule	Implemented	Wired	Attestation primitives	Not-invented
security	i18n	Submodule	Implemented	Wired	i18n	Not-invented

vision_engine/

Component	Feature	Category	Implementation	Wiring	Real-use
vision_engine	Analyzer interface	Submodule	Implemented	Wired	UI-element/text-region, issue detection, screen diff, key frames
vision_engine	LLM vision adapters	Submodule	Implemented	Wired	OpenAI GPT-4o, Anthr Claude, Gemini 2.0 Fla Qwen VL, Kimi, StepF Ollama + Fallback chain
vision_engine	NavigationGraph	Submodule	Implemented	Wired	Directed graph of screen transitions
vision_engine	Remote Ollama deployment	Submodule	Implemented	Wired	SSH auto-install, mode lifecycle; GPU/CPU/R detection; distributed IL RPC
vision_engine	OpenCV (stub + build-tag real)	Submodule	Partial	Wired	OpenCV-based CV — s default, real impl behind tag
vision_engine	Config / i18n	Submodule	Implemented	Wired	Configuration (HELIX_VISION_PRO etc.), i18n

Honest gaps & caveats (§11.4.6)

- Video coverage is 30 feature rows.** 30 features are confirmed by real, ffprobe- and content-verified recordings (Wave 2 of 2026-06-16 added 5 net-new — DOCX output, PDF output, HTML→Markdown conversion, ProviderDiverseTranslators, EnsembleFactory — plus an additional re-confirming clip on the already-confirmed gRPC TranslationService row, no double-count; Wave 1 added 2 — bridge-ensemble best-model selection + bridge provider-diverse catalogue; see the Anti-bluff note + the Coverage-

- summary Video-confirmed row for the authoritative count and full list — reconciled with both, §11.4.60). DOCX output and PDF output, formerly real-artifact-only, are now video-confirmed by Wave-2 recordings. Every other runtime/user-visible feature is PENDING a recording or N/A. This is stated, not hidden. 1a. **DOCX output is Implemented (Rev 3, commit 87cd2be)** — pure-Go OOXML writer; a real run produced a Microsoft Word 2007+ .docx; **now video-confirmed (Wave 2)** by helix_translate-docx-output-20260616-112943.mp4 (openai/gpt-4, real Spanish in the .docx). **PDF output is Implemented (Rev 4, commit fb265e7)** — pkg/ebook/pdf_writer.go (Book→HTML5→weasyprint→Cyrillic-faithful PDF), wired into unified-translator (.pdf output, main.go:1160); pkg/ebook PDF tests green; **now video-confirmed (Wave 2)** by helix_translate-pdf-output-20260616-113057.mp4 (openai/gpt-4, valid PDF with real Spanish). Output formats are EPUB/FB2/TXT/HTML/MD/DOCX/PDF. 1b. **-multipass ≠ -verify**. The unified-translator -verify flag runs the CLI's own per-step output check; the separate -multipass flag (d53e085) invokes the pkg/verification multi-pass polisher engine — now wired and video-confirmed. 1c. **cmd/translator, cmd/translate-ssh, cmd/ebook-translator are now Obsolete (→ Fixed.md), Rev 7** — RECONCILED with the per-binary table rows (lines ~136/218/233) which mark all three Obsolete (→ Fixed.md) with Reason=feature-removed. The local-runtime + SSH-local translation paths were REMOVED in bridge phase-2 R-2..R-4 (operator-approved 2026-06-15 D1/D2 + R-4 “keep distributed API, remove SSH-local”); the default path now sources the LLMsVerifier bridge (pkg/bridge). These are therefore NO LONGER open STUB / OPERATOR-BLOCKED items — the earlier “STUB / OPERATOR-BLOCKED” wording was stale and is corrected here. The KEPT distributed-API translation path is NOT Obsolete. 1e. **GET /api/v1/verified-models returns 404** — LLMsVerifier is disabled in the current config so the route is not mounted (feature-disabled-by-config, not a defect). 1f. **cmd/cli title path-leak FIXED** (d53e085) + **write-safety FIXED** (87cd2be, atomic write/no-clobber, guard cmd/cli/no_partial_output_test.go).
2. **Tests = Not-inventoried** for all rows: the raw inventory did not perform per-feature test attribution. The repo carries extensive _test.go + test/ suites, but mapping each to a feature is future work — recorded as honest unknown, never guessed Covered/None.
 3. **4 STUB endpoints** return mock data and do NOT do the named work: POST /api/v1/translate/ebook, POST /api/v1/preparation/analyze, GET /api/v1/preparation/result/:id, POST /api/v1/translate-with-verification.
 4. **1 Partial endpoint:** standalone pkg/api/server.go POST /api/batch returns a queued batch_id only and performs no translation.
 5. **Static/hardcoded returns** (not stubs, but not live): GET /api/stats (zeros), GET /api/v1/status/:id (hardcoded completed), GET /api/v1/metrics (static), monitor-server GET /api/v1/status (static monitoring_active), several static language/provider lists. Marked Implemented with the static behaviour stated in Real-use.
 6. **vision_engine OpenCV** is Partial — stub by default, real CV only behind a build tag.
 7. **UNCONFIRMED LLM determinations** carried verbatim from the inventory for thin OpenAI-compatible providers, several llms_verifier packages, challenges/userflow evaluators, and doc_processor/archdoc — noted inline rather than asserted.
 8.  **Wave-1 DISCOVERED BLOCKER — RESOLVED (Wave 2, §11.4.4 / §11.4.107 / §11.4.108 / §11.4.138).** The default bridge TRANSLATOR-path

runtime failure described below is FIXED: the Wave-2 recordings (DOCX output, PDF output, gRPC translate, and the 4 per-provider unified clips novita/mistral/groq/siliconflow) are real-use videos validator-verified (ffprobe+frame) showing the verified-bridge translator path producing real target-language output — the exact path that failed in Wave 1 now works, which is why those rows flip to CONFIRMED this wave. The historical forensic record of the blocker is preserved below for §11.4.6 / §11.4.138 traceability. **[HISTORICAL — Wave-1 finding, now fixed]** While recording Wave 1 (DOCX/PDF/gRPC), a real run of unified-translator (and, by source, the gRPC backend pkg/grpc/core_translator.go which uses BestTranslatorFunc) FAILS at translate time with:

```
bridge: no verified translator available: model 'Sao10K/L3-8B-Stheno-v3.2' is not valid for provider 'novita'. Valid models: [meta-llama/llama-3.1-70b-instruct mistralai/mixtral-8x7b-instruct]. Root cause (FACT, §11.4.6):
```

VerifiedFactory.CreateTranslatorWithFallback / CreateTranslator build the verified translator via NewLLMTranslatorWithConfig, which re-validates the model against the STALE hardcoded ValidModels whitelist (pkg/translator/llm/llm.go:227-241); a genuinely LLMsVerifier-verified model whose ID is not in that whitelist is rejected. The BestModel/invoke path (used by model-bridge invoke + Wave-1 videos) correctly BYPASSES this whitelist (its own code comment, bridge.go:250-258, documents exactly why a verified model must not be gated on the whitelist) — and so does NOT exhibit the bug. **Impact:** unified-translator (all outputs: TXT/HTML/EPUB/FB2/MD/DOCX/PDF) + gRPC translate are non-functional via the default bridge path until fixed. This BLOCKS Wave-1 videos #2 (DOCX), #3 (PDF), #4 (gRPC), and the per-provider unified-path; recording any of them as “working” would be a §11.4.107 bluff, so they are honest SKIPS this wave. **Fix direction (not yet applied — needs its own §11.4.115 RED→GREEN + §11.4.142 independent review + §11.4.145 impact research + §11.4.40 full-suite retest):** route VerifiedFactory’s verified-translator construction through the ResolvedProvider resolver (FactoryProvider=“openai” + BaseURL) the way bridge.realClientBuild already does, bypassing the ValidModels whitelist for already-verified models. Tracked as a Wave-2 / dedicated fix item (severity HIGH — release-critical default path).